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Ion microprobe U–Pb ages for Neoproterozoic basaltic magmatism in south-central Australia and implications for the breakup of Rodinia

Michael T.D. Wingate [a,*], Ian H. Campbell [a], William Compston [a], George M. Gibson [b]

[a] *Research School of Earth Sciences, Australian National University, Canberra, ACT 0200, Australia*
[b] *Australian Geological Survey Organization, Canberra, ACT 2601, Australia*

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Abstract

Ion microprobe U–Th–Pb analyses of baddeleyite and zircon yield precise estimates of the age of Neoproterozoic mafic magmatism associated with initial rifting in the Adelaide Geosyncline and adjacent platforms in southern and central Australia. An age of 827 ± 6 Ma is inferred for emplacement of the Gairdner Dyke Swarm (GDS), based on 137 $^{207}Pb/^{206}Pb$ analyses of 66 baddeleyite crystals from a drill-core sample of GDS dolerite. It has been proposed elsewhere that the GDS, together with ~ 780 Ma mafic intrusive suites in three widely separated areas of western North America, represent sectors of a giant radiating dyke swarm fragmented by the separation of Australia and Laurentia during breakup of the Rodinia supercontinent. Our results show that the Australian dykes are at least 40 Ma older than the intrusions of North America and therefore cannot have been emplaced during the same event. The Little Broken Hill gabbro and associated dykes, situated close to the inferred Neoproterozoic margin of Australia, have been linked previously with extension related to breakup between Australia and Laurentia. If correct, then a zircon $^{207}Pb/^{206}Pb$ age of 827 ± 9 Ma obtained for the gabbro, coupled with evidence that the GDS is plume-related rather than breakup-related, indicates that breakup commenced soon after emplacement of the GDS at 827 Ma and that the mantle plume is likely to have played an active role in triggering the process. Breakdown of primary baddeleyite in the Little Broken Hill gabbro to form overgrowths of polycrystalline zircon occurred at 500 ± 7 Ma, thereby providing a precise estimate of the age of the thermal event associated with the Delamerian Orogeny in this region. Our results show the GDS and related rocks in Australia to be roughly coeval with mafic igneous rocks in similar stratigraphic position in the Lower Sinian System and may lend support to the hypothesis that parts of South China were situated between Australia and Laurentia prior to the breakup of Rodinia. © 1998 Elsevier Science B.V.

Keywords: Australia; Baddeleyite; Dyke; Neoproterozoic; Rodinia; Zircon

Corresponding author. Present address: Tectonics Special Research Centre, Department of Geology and Geophysics, University of Western Australia, Nedlands, WA 6907, Australia. Fax: +61 8 9380 1090; e-mail: mwingate@geol.uwa.edu.au

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1. Introduction

The hypothesis that Australia and Antarctica were joined to western Laurentia (Jefferson, 1978; Bell and Jefferson, 1987) is an integral part of reconstructions (Dalziel, 1991, 1992; Hoffman, 1991; Moores, 1991; Ross et al., 1992; Brookfield, 1993; Powell et al., 1993; Borg and DePaolo, 1994) of the late Proterozoic supercontinent Rodinia (McMenamin and Schulte-McMenamin, 1990). Although much evidence has been marshalled in support of an Australia–Laurentia connection during the late Proterozoic, the hypothesis has yet to be tested rigorously. The association of major dyke swarms with continental fragmentation and their potential utility as time-rock markers for elucidating and testing continental reconstructions is now well recognized (May, 1971; Morgan, 1981; Fahrig, 1987; Buchan and Halls, 1990; Heaman et al., 1992; Dalziel and Lawver, 1995; Ernst et al., 1995; Park et al., 1995; Wingate et al., 1995).

The breakup of continental crust is commonly associated with the arrival of a new mantle plume (Morgan, 1981). Where this happens, a continental flood basalt (CFB) province forms which may become separated into two parts when the continent divideş. Well-studied examples include the Parana–Etendeka province, associated with the opening of the South Atlantic (Peate et al., 1990) and the Tertiary province of Britain and Greenland, related to opening of the North Atlantic (White and McKenzie, 1989). In CFB provinces of Precambrian age, most basalts are likely to be lost by erosion, although areally extensive swarms of mafic dykes that acted as feeders to the basalts, are typically well-preserved. Dyke swarms can be oriented in a radial pattern in the stress field above a rising plume (May, 1971; Fahrig, 1987; Ernst et al., 1993, Ernst et al., 1995), or can follow pre-existing crustal weaknesses. Alternatively, during continental breakup, dykes may be emplaced in coast-parallel swarms along each new continental margin and, in some cases, along a failed rift or aulacogen (Burke and Dewey, 1973; Nielsen, 1978; Hooper, 1990; Peate et al., 1990). The time scales of both flood volcanism and continental breakup are short, typically less than 10 Ma (White and McKenzie, 1989; Hill et al., 1992); hence these dyke swarms afford two mechanisms for testing hypotheses regarding the original positions of continents and the kinematics of their breakup. First, if dyke swarms on now separated continents once formed a conjugate pair, their ages should be the same within 10 Ma. Second, in a prebreakup reconstruction, the conjugate dykes should share a common paleomagnetic pole.

Park et al. (1995) presented geochronological and paleomagnetic results from three 780 Ma mafic intrusive suites in western North America. Within the framework of the Rodinia reconstructions, they suggested that these intrusions, together with the Gairdner Dyke Swarm (GDS) of south-central Australia (Fig. 1), represent sectors of a giant radiating dyke swarm emplaced above a mantle plume prior to continental separation. If the GDS could be shown to be the same age, within 5 or 10 Ma as the North American intrusions, this would constitute very strong evidence in support of an Australia–Laurentia connection in the late Proterozoic. However, Sm–Nd mineral isochron ages for the GDS (Zhao and McCulloch, 1993) are insufficiently precise to confirm its equivalence to the intrusions of western North America. Without a precise and unambiguous age for the GDS and related mafic igneous rocks, the timing of associated rifting and subsequent development of the Pacific Ocean basin will remain controversial issues (Preiss, 1987, 1990; Drexel et al., 1993; Powell et al., 1994; Dalziel, 1997).

This study was designed initially to determine precise U–Pb ages for the GDS, to establish if it is equivalent in age to the North American intrusions. We expanded our work to investigate the ages of mafic intrusions in the Broken Hill area that are inferred, from recent structural and geochemical information (Gibson et al., 1996, Gibson et al., 1997), to have been emplaced during Neoproterozoic extension. In this paper, we present ion microprobe U–Th–Pb data for baddeleyite and zircon that provide precise estimates of the age of initial rift-related basaltic magmatism in south-central Australia. These results resolve controversy regarding the age of this event and have implications for hypotheses regarding the original configuration and subsequent breakup of Australia, Laurentia and South China. As these

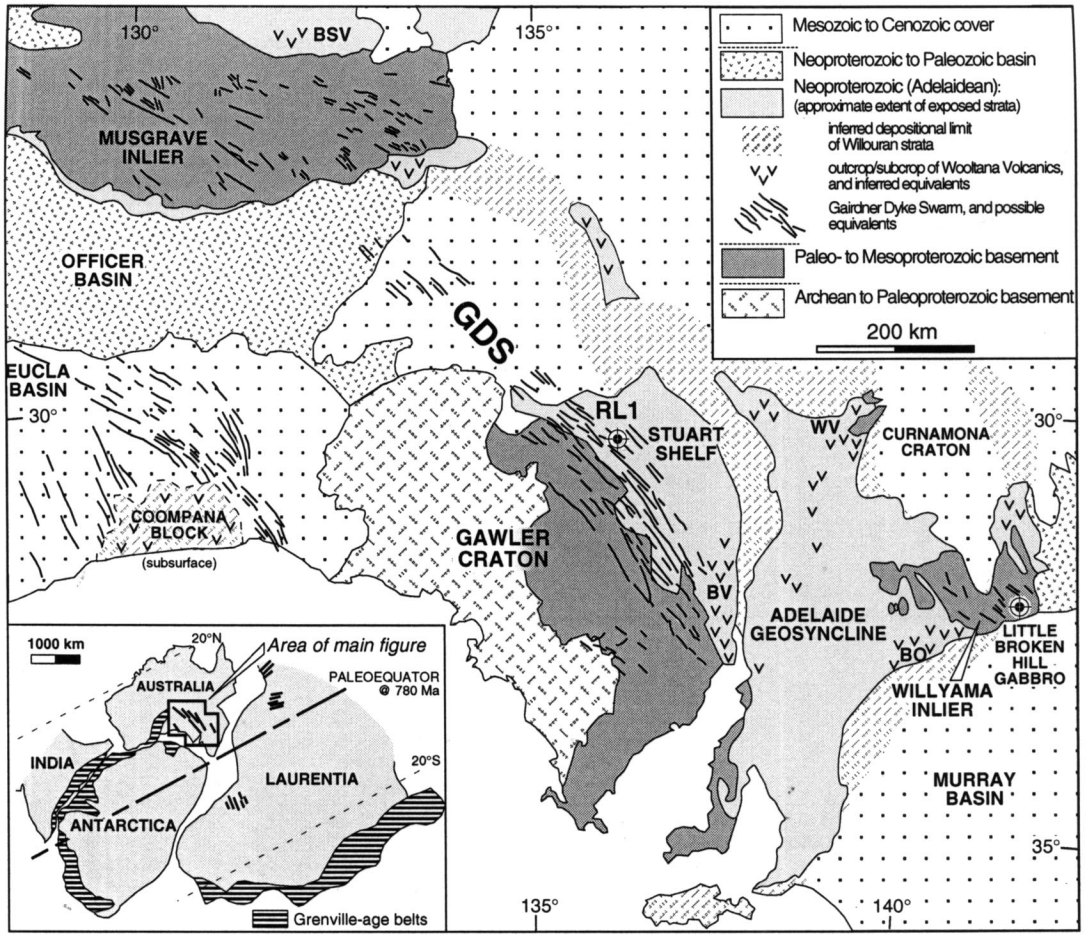


Fig. 1. Simplified geological map showing the Gairdner Dyke Swarm (GDS) and associated mafic igneous rocks in relation to main tectonic provinces of south-central Australia, compiled from Stroud et al. (1983), Preiss (1987), Boyd and Tucker (1990) and Drexel et al. (1993). Also shown are sample locations, RL1 and Little Broken Hill gabbro, for rocks dated during this study. Rock units referred to in the text include: WV, Wooltana Volcanics; BO, Boucaut Volcanics, BV, Beda Volcanics; BSV, Bitter Springs Volcanics. Only the dykes in the Musgrave and Willyama Inliers are exposed in outcrop; others are known from aeromagnetic data and/or drill hole samples only. Subsurface mafic dykes and volcanics of the Coompana Block are possible GDS equivalents (Flint and Daly, 1993), but have not been dated. The inset shows the Neoproterozoic Rodinia reconstruction of Borg and DePaolo (1994) with 780 Ma Laurentian mafic intrusions and paleolatitudes of Park et al. (1995).

are the first published analyses of baddeleyite using the SHRIMP ion microprobes, details of calibration and correction procedures are provided.

2. Geology and previous geochronology

2.1. The Adelaide Geosyncline and Gairdner Dyke Swarm

The Adelaide Geosyncline is a complex of extensive, deeply subsided sedimentary basins of Neoproterozoic to Cambrian age that overlies Paleoproterozoic to Mesoproterozoic basement in south-central Australia (Preiss, 1987). Sedimentary sequences of the Adelaide Geosyncline (Fig. 2) have been correlated with similar packages in western North America (Jefferson, 1978; Bell and Jefferson, 1987; Ross et al., 1992; Young, 1992) and in the Sinian Basin of South China (Eisbacher, 1985; Li et al., 1995, 1996). The Wooltana Volcanics near the base of the Adelaide Geosyncline are a

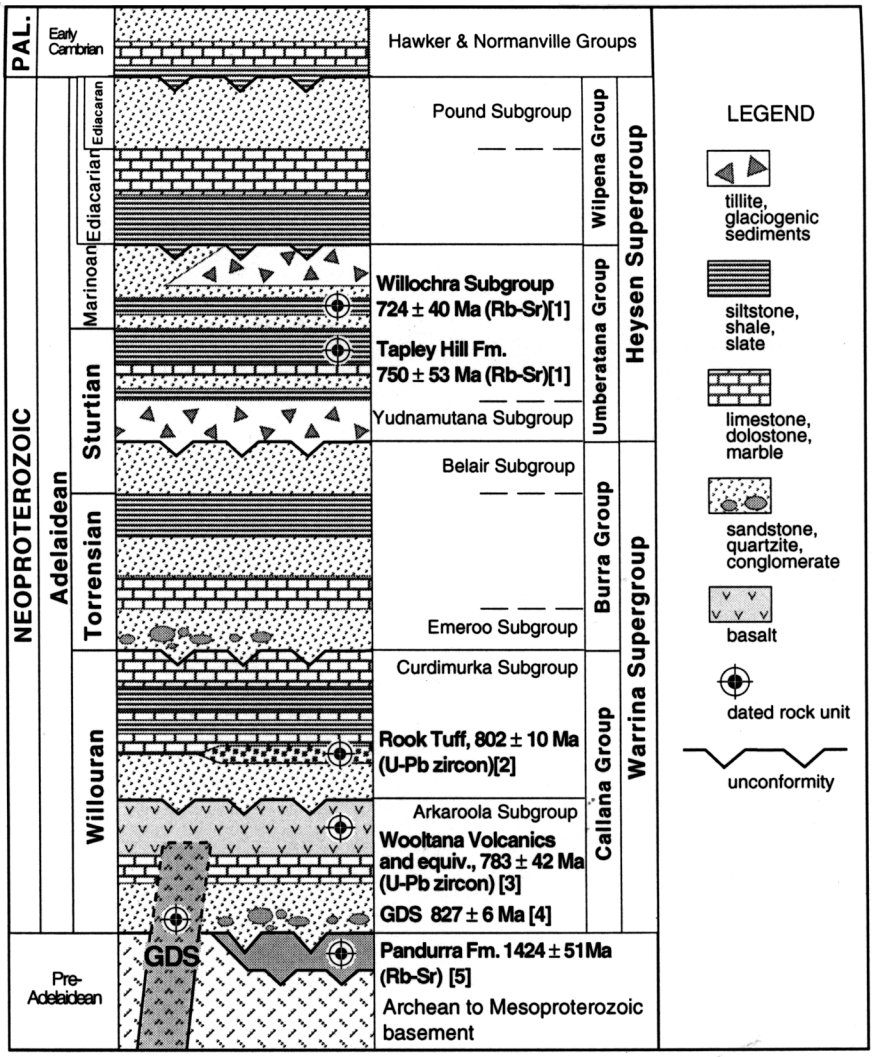


Fig. 2. Schematic composite cross-section for Neoproterozoic rocks of the Adelaide Geosyncline and adjacent platforms showing chronostratigraphic and selected lithostratigraphic units (simplified from Preiss, 1987; Drexel et al., 1993). No thicknesses are implied and the entire sequence is not present at any one locality. Dated rock units are described in detail in the text. References: [1] Preiss (1987); [2] Fanning et al. (1986); [3] Fanning et al. (1989), in Drexel et al. (1993); [4] this study; [5] Fanning et al. (1983).

$\leq$2 km thick sequence of sub-aerial tholeiitic continental flood basalts that record extension and rifting during the initial stages of basin formation that preceded continental breakup (Preiss, 1987, 1990; Hilyard, 1990; Crawford and Hilyard, 1990; Powell et al., 1994). The Wooltana Volcanics have stratigraphic equivalents throughout the Adelaide Geosyncline, on adjacent platforms of the Gawler Craton (Stuart Shelf) and Cumamona Craton, and beneath younger rocks in surrounding basins (Preiss, 1987, 1990, 1993; Hilyard, 1990). Most rocks within the Adelaide Geosyncline were strongly deformed and metamorphosed during the late Cambro-Ordovician Delamerian Orogeny, although adjacent platformal strata were less affected (Preiss, 1990).

The GDS is recognized on aeromagnetic images as a NW-trending array of linear magnetic anomalies (Tucker and Boyd, 1987; Boyd and Tucker, 1990), most prominent on the Stuart Shelf, where the dykes are buried beneath Sturtian and younger strata (Preiss, 1987, Preiss, 1990). Individual dolerite dykes have been intersected in several drill holes (Preiss, 1983) and are inferred to be subvertical, to range from ~30 to 150 km in length and probably to exceed 50 m in width (Cowley and Flint, 1993). The orientation of the dyke swarm is consistent with intrusion during NE–SW-directed extension.

The GDS was equated with similarly trending dykes in the Musgrave Inlier by Goode (1970), indicating a total strike length of ~1000 km (Fig. 1). This correlation is supported by strong similarities in geochemical and isotopic data that imply that the GDS, the Wooltana Volcanics (and equivalents within the Adelaide Geosyncline), the Beda Volcanics of the Stuart Shelf, dolerite dykes of the Musgrave Inlier, and the Bitter Springs Volcanics of the Amadeus Basin and Arunta Inlier could be cogenetic (Giles and Teale, 1979; Crawford and Hilyard, 1990; Zhao and McCulloch, 1993; Zhao et al., 1994). Dolerite and gabbro components are observed within the Wooltana Volcanics and are common xenoclasts within diapiric breecias of the Adelaide Geosyncline (Preiss, 1987; Hilyard, 1990). Relatively undeformed basalts, NW-trending dykes and gabbroic plutons occur in the Willyama Inlier (Stroud et al., 1983); intrusive rocks have been correlated tentatively with the GDS (Flint and Parker, 1993). Collectively these rocks extend over an area of ~210 000 km^2 and were defined as the Willouran Basic Province by Crawford and Hilyard (1990), who showed that geochemical and other characteristics are similar to those of several younger CFB provinces. Zhao et al. (1994) argued that trace element and Nd isotopic signatures of the GDS and associated volcanic rocks are consistent with their derivation by decompression melting of a mantle plume head.

If the correlations described above are correct, then on stratigraphic grounds, the GDS was emplaced between 1424±51 Ma, an Rb–Sr isochron age for the Pandurra Formation, the youngest intruded pre-Adelaidian basement (Fanning et al., 1983) and 802±10 Ma, a concordant U–Pb zircon age for the thin, dacitic Rook Tuff (Fig. 2) within upper Willouran sediments that overlie the Wooltana basalt (Fanning et al., 1986). (All ages are quoted with either 2σ or 95% confidence limits and Rb–Sr ages, where necessary, have been recalculated using $\lambda_{87_{Rb}} \times 1.42 \times 10^{-11}\,a^{-1}$.) Dykes of the GDS do not intrude Torrensian or younger rocks (Fig. 2), indicating that they are of Willouran age or older (Hilyard, 1990; Crawford and Hilyard, 1990).

Attempts to date directly the GDS and associated volcanic rocks have produced equivocal results. Dolerite from drill hole Shell Reedy Lagoon DDH1 (RL1) yielded a minimum K–Ar age of 733±7 Ma (Cowley and Flint, 1993) and an imprecise Rb–Sr age of 1132 Ma (Fanning, 1984, Preiss, 1987). Possible GDS dolerite from drill hole RD 222 at Olympic Dam gave an Rb–Sr age of 1070±74 (R. Creaser, pers. comm. in Drexel et al., 1993). A minimum age of 830±50 Ma was inferred for eruption of the Wooltana Volcanics by Compston et al. (1966), based on a Rb–Sr isochron and whole-rock Rb–Sr ages of 697±70 Ma (Webb and Horr, 1978) and 1076±34 Ma (Webb and Coats, 1980, Preiss, 1987) were obtained for the Beda Volcanics. A U–Pb zircon upper intercept age of 783±42 Ma was indicated for eruption of rhyolite of the Boucaut Volcanics, located SW of the Willyama Inlier, based on six discordant conventional analyses (Fanning, 1989 in Drexel et al., 1993).

More recently, apparent ages of 867±47 and 802±35 Ma, based on Sm–Nd mineral isochrons, were obtained for GDS dolerite from RL1 and Aquitaine drill hole SSR 1001, respectively, by Zhao and McCulloch (1993). These authors also reported Sm–Nd dates of 790±40 and 797±47 Ma for NW-trending dykes near the town of Amata in the eastern Musgrave Inlier. They inferred that the GDS was emplaced between 790 and 870 Ma. Zircon extracted from a NW-trending dolerite dyke in the western Musgrave Inlier yielded a $^{207}Pb/^{206}Pb^*$ age of 824±4 Ma (Glikson et al., 1996).

2.2. The Willyama Inlier and Little Broken Hill gabbro

The Willyama Inlier, on the north-eastern margin of the Adelaide Geosyncline (Fig. 1) consists of strongly deformed metamorphic basement of Paleoproterozoic age, intruded by mafic dykes and plutons and overlain unconformably by Adelaidean and younger strata. The latter are preserved in NW-trending rift basins and half-grabens on the inlier and overlie its margins (Preiss, 1990; Flint and Parker, 1993). Basement and cover rocks were deformed during Neoproterozoic rifting and during Cambro-Ordovician (Delamerian) tectonism. Based on structural mapping aided by new aeromagnetic and geochemical data, Gibson et al. (1996, 1997) argued that two separate mafic intrusive suites intruded mid-Proterozoic basement in the Broken Hill region (cf. Stroud et al., 1983), although their ages are not well constrained.

A younger suite, possibly related to the Delamerian Orogeny, includes NW-trending dolerite and pyroxenite dykes. A K–Ar age of 561 ± 14 Ma was obtained by Harrison and McDougall (1981) for hornblende from a nepheline pyroxenite intrusion that probably belongs to the younger suite. They interpreted this as the age of emplacement, although K–Ar and Rb–Sr isotopic systems in many minerals and rocks in the region were reset by a thermal pulse at ~ 520 Ma (Harrison and McDougall, 1981 and references therein).

The Little Broken Hill gabbro belongs to the older suite, which includes minor uralitic gabbro plutons and various dykes that in some cases, were intruded into and parallel to retrograde shear zones interpreted to have been active extensional structures during Neoproterozoic rifting (Gibson et al., 1996, 1997). Recent geochemical data and detailed geological mapping indicate that the Little Broken Hill gabbro represents a magma chamber that fed dykes exposed at structurally higher levels, including NW-trending dykes intruded into and parallel to the Stirling Hill shear zone (Gibson et al., 1996, 1997). NW-trending uralitic dykes in the western part of the inlier are overlain unconformably by lower Adelaidean strata, suggesting their emplacement during initial rifting in the Adelaide Geosyncline (Flint and Parker, 1993).

3. Sample descriptions

3.1. Gairdner Dyke Swarm, Reedy Lagoon 1 (RL1)

Core samples were obtained from several drill holes that intersected dolerite of the GDS, but only a medium-grained (2 to 3 mm) dolerite sample (671 to 673 m depth) from RL1 (30.32°S, 136.07°E) contained baddeleyite and zircon. Microscopic examination reveals mainly calcic plagioclase and sub-ophitic clinopyroxene (augite), with interstitial magnetite, ilmenite and quartz and trace amounts of apatite, baddeleyite and zircon. Augite exhibits slight to moderate alteration to actinolite and chlorite.

Using standard magnetic and density separation techniques, 60 zircon and ~ 200 baddeleyite crystals and crystal fragments were recovered from ~ 750 g of crushed sample. Baddeleyite crystals are euhedral and tabular, inclusion-free, and range up to 150 μm in length. Concentric euhedral zoning is visible in most crystals using cathodoluminescence imaging. Many crystals contain irregular or vermiform cavities and embayments (Fig. 3) that might be primary growth features, or else might reflect partial resorption via late-stage, silica-enriched fluids. Most RL1 zircons are euhedral, equant to elongate (aspect ratios up to 5:1), between ~ 80 and ~ 250 μm in length and exhibit concentric growth zoning. Irregular cavities, similar to those observed in baddeleyite, are common; other zircons have a distinctly skeletal appearance and possess central channels (Fig. 3). These textures, in both zircon and baddeleyite, are indicative of very rapid crystallization (e.g. Lofgren, 1980). Quartz, apatite and a ferromagnesian aluminosilicate (orthopyroxene?) were identified by electron probe within both cavities and channels.

RL1 zircons exhibit variable degrees of metamictization. Almost all grains show some discolouration; many are strongly discoloured and/or possess areas opaque to transmitted light and show abundant cracks. Cathodoluminescence imaging shows

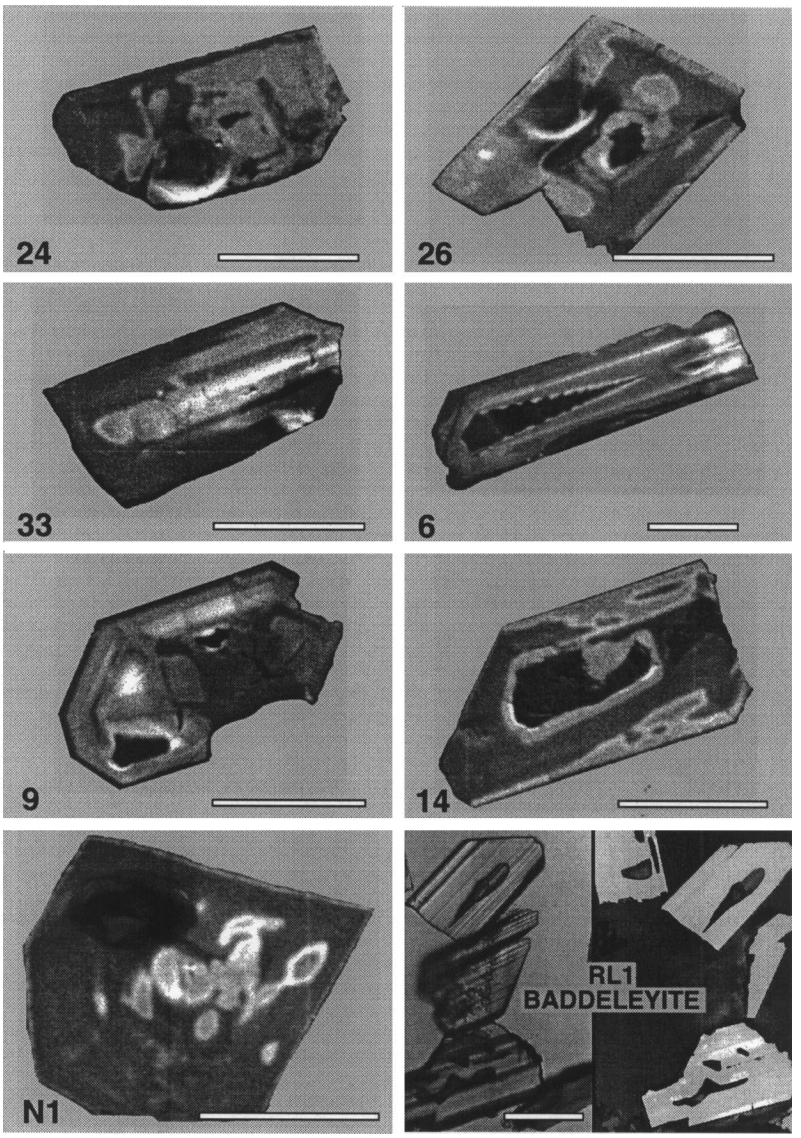


Fig. 3. Cathodoluminescence images illustrating the distribution of luminescent and non-luminescent areas in zircons from sample RL1. Based on these and other images, we interpret the normally luminescent zircon to be original and non-luminescent zircon as the result of alteration which enriched these areas in U and Th (thereby inhibiting the cathodoluminescence effect). Numbers refer to analyses in Table 1; scale bars are 50 μm. Oval depressions are SHRIMP analysis pits from session 2. Cavities in crystals 26, 9, and 14 and the central channel in crystal 6, contain quartz, apatite and other silicate minerals. The tapered appearance of light and dark areas in crystal 33 and of the channel in crystal 6 is due to these crystals having been polished at an angle slightly oblique to the *c*-axis. Crystal N1 was not dated; the SHRIMP pit resulted from scanning the ^{204}Pb peak to assess possible isobaric interference. The image in the lower right consists of transmitted (left) and reflected light (right) photomicrographs, showing irregular cavities common in RL1 baddeleyite.

that many crystals contain areas of normally luminescent zircon surrounded or transected by thick and irregular, non-luminescent rims and internal domains (Fig. 3). The lack of cathodoluminescence in the latter areas is consistent with radiation damage due to high concentrations of U and Th

(verified by SHRIMP analysis, as described below). Although relationships between these two zircon types in many crystals are ambiguous, we conclude, based on examination of most of the crystals in the sample that the luminescent zircon is original and that high-U, non-luminescent areas resulted from some form of alteration. In general, the non-luminescent areas do not coincide with the dark areas seen in transmitted light. Many zircons possess areas that are both normally luminescent as well as clear and colourless in transmitted light; it was anticipated that analysis of these areas would yield the most consistent age results.

3.2. Little Broken Hill gabbro (96180040)

Surface samples were collected from several mafic intrusions in the eastern Willyama Inlier; a medium- to coarse-grained gabbro from Little Broken Hill (32.10°S, 141.54°E) was found to contain zircon and baddeleyite. The sample is igneous-textured, but is extensively uralitized and is seen in thin section to consist of approximately equal parts of secondary amphibole and distinctively brown plagioclase. Primary pyroxene (noted in other samples, Stroud et al., 1983) and plagioclase are wholly or partially replaced by a metamorphic assemblage of colourless to blue–green amphibole, biotite, quartz, titanite, chlorite and clinozoisite. Accessory minerals include apatite, magnetite and pyrite.

More than 100 subhedral to euhedral zircons were separated from 600 g of sample. They range in size up to ~200 μm and most are clear, colourless and free of inclusions. Cathodoluminescence imaging reveals few internal structures; a small minority display euhedral growth zones. The sample also yielded 10 anhedral to subhedral brownish, translucent grains with darker cores. In cathodoluminescence images, the thick rims luminesce weakly, while the cores remain dark. Using the SEM in energy-dispersive mode, the cores were shown to be baddeleyite and the overgrowths to consist of polycrystalline zircon.

4. Analytical procedures

Each sample, together with calibration standards, was cast in an epoxy mount, then polished to expose the crystals' interiors. Polished mounts were documented with transmitted and reflected light micrographs and cathodoluminescence images. Each mount was then cleaned thoroughly and vacuum-coated with gold. To eliminate residual hydrides from sample surfaces, mounts were stored for >20 h in a 60°C oven, then pumped down to high vacuum overnight in the SHRIMP sample lock prior to analysis.

Measurements of U, Th and Pb in zircon were conducted using the SHRIMP ion microprobes at the Research School of Earth Sciences, using standard operating and data processing procedures (Compston et al., 1984, 1992; Claoue-Long et al., 1995). As found for zircon, Pb isotope ratios measured during SHRIMP analysis of baddeleyite are constant nearly to within counting statistics and, based on comparisons with baddeleyite analysed by isotope dilution thermal ionization mass spectrometry (Fig. 4), no correction for isotope discrimination is applied to baddeleyite results. As pointed out previously (Compston et al., 1996), any bias in $^{207}Pb^*/^{206}Pb^*$ (Pb^* denotes radiogenic Pb) due to sputtering, which enhances the light isotopes (Shimizu and Hart, 1982), would be obscured by the possible presence of $^{206}Pb/^{1}H$. Although no direct monitoring of PbH^+ is practical for baddeleyite, plots of measured $^{207}Pb^*/^{206}Pb^*$ versus time for each session are quite constant rather than slowly decreasing as vacuum is improved, indicating that the presence of PbH^+ can be discounted for these analyses.

Variable discrimination in $^{206}Pb^+/^{238}U^+$ is corrected using an observed power law variation with UO^+/U^+ in a manner similar to that described previously for zircon (Compston et al., 1984, Compston et al., 1992; Claoue-Long et al., 1995). Bivariate regression analysis of 37 data sets (Fig. 5) was employed to determine a characteristic power law exponent for baddeleyite of 2.12 ± 0.02 (1σ), similar to results for zircon (2.0, Claoue-Long et al., 1995). However, $^{206}Pb^*/^{238}U$ ratios measured in baddeleyite by ion microprobe are found consistently to be dispersed well beyond analytical precision. Experiments using baddeleyite crystals arranged in particular orientations reveal that measured $^{206}Pb^*/^{238}U$ ratios vary significantly (up to 10% or more) and systematically with the relative

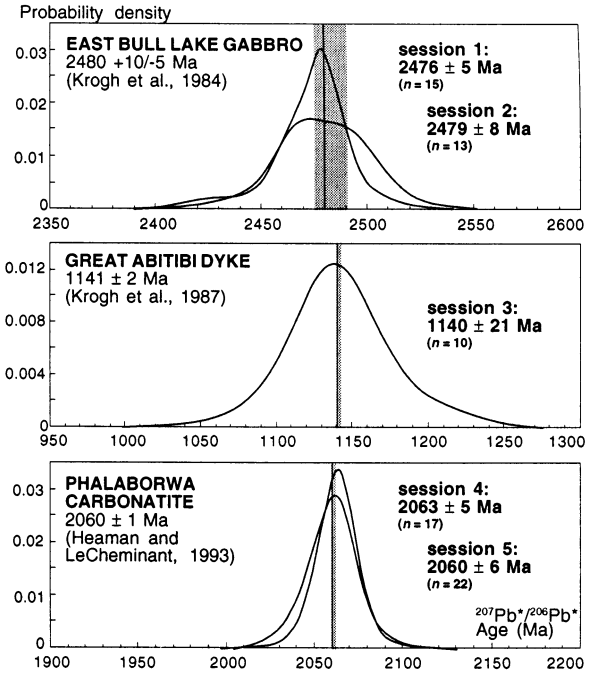


Fig. 4. Comparison of $^{207}Pb^*/^{206}Pb^*$ ages measured by SHRIMP (probability density curves) with published isotope dilution measurements (shaded areas illustrate error limits) for three baddeleyite reference samples analysed concurrently with RL1 baddeleyite, Krogh et al., 1984, Krogh et al., 1987 and Heaman and LeCheminant, 1993. In each of five sessions, ion microprobe and conventional results agree to within analytical precision, demonstrating the accuracy of $^{207}Pb^*/^{206}Pb^*$ ages measured in baddeleyite by ion microprobe. Age uncertainties are ±95%; *n*, number of analyses. The probability density curves (Silverman, 1986) are standardized so that the area under each sums to a probability of 1; vertical scales indicate the probability of finding an event within a 1 Ma interval; the probability of finding an event in larger intervals is given by the area under that portion of the curve.

orientation of the baddeleyite crystal structure and the direction of the primary ion beam (Wingate, 1997). Owing to this effect, we report here only $^{207}Pb^*/^{206}Pb^*$ ratios, which are unaffected by orientation (Wingate, 1997).

Correction of measured compositions for common Pb was carried out differently for each sample and is described below. Correction for common Pb in most zircons employed the measured ^{204}Pb. For baddeleyite and low $^{232}ThH/^{238}U$ zircon, common Pb was subtracted from measured compositions using the '208-method'

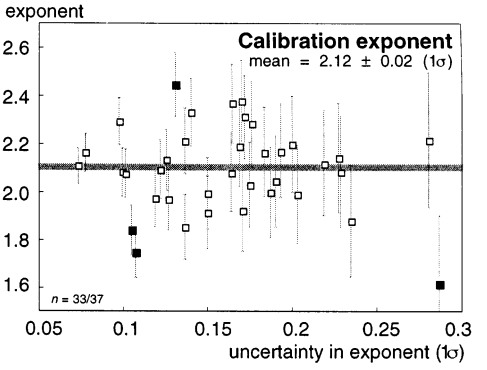


Fig. 5. Values of the power law calibration exponent for baddeleyite, plotted against the uncertainties in their determination. The weighted mean value used in subsequent data reduction is based on 33 of 37 data sets; four outlying values are shown with closed symbols.

(Hinthorne et al., 1979; Compston et al., 1984), whereby the proportion of common Pb is estimated from the difference between measured $^{208}Pb/^{206}Pb$ and the radiogenic ratio expected for the age and $^{232}Th/^{238}U$ of the mineral. In all cases, corrections are sufficiently small to be insensitive to the choice of common Pb composition and an average crustal composition appropriate to the age of the mineral was assumed. Prior to analysis, each site was cleaned thoroughly by rastering the primary ion beam over the area for 3 min. Subsequently, ^{204}Pb counts for most analyses remained low and constant and showed no tendency to decrease over the course of a 20 min analysis, indicating that common Pb in these samples is mainly inherent to the mineral rather than surface-related.

On the basis of the excellent correlation observed between $^{208}Pb^*/^{206}Pb^*$, and ThO^+/UO^+, ratios of $^{232}Th/^{238}U$ in baddeleyite are found to be linearly proportional to measured ThO^+/UO^+ and insensitive to variable discrimination during sputtering as measured by UO^+/U^+. Fifteen data sets from three samples of known age were employed to determine the discrimination factor required to correct ThO^+/UO^+ to the value of $^{232}Th/^{238}U$ expected for measured $^{208}Pb^*/^{206}Pb^*$ (corrected independently for common Pb using the 207-method: Compston et al., 1992, 1995; Roberts et al., 1995). The very low $^{232}Th/^{238}U$ typical of

baddeleyite means that radiogenic $^{208}Pb^*/^{206}Pb^*$ ratios are correspondingly low and hence, to ensure that only reliable results were employed, only those data were considered for which ThO^+/UO^+ was greater than 0.01 and the proportion of common to total ^{206}Pb was less than 0.01 (Fig. 6). The value obtained of 1.09 ± 0.01 (1σ) is similar to the value of 1.11 derived previously for zircon (Compston et al., 1984). It should be noted that owing to the low common Pb content of baddeleyite, the contribution from uncertainty in the $^{232}Th/^{238}U$ discrimination factor to final errors in radiogenic ratios is negligible.

During this study, approximate values for U concentration in baddeleyite were determined using the same power-law relationship between Zr_2O^+/U^+ and UO^+/U^+ employed for zircon (Claoue-Long et al., 1995), in which the value of the power law exponent is 0.66. Similarity in the power-law calibration of $^{206}Pb^*/^{238}U$ for baddeleyite and zircon indicates that the relationship between UO^+/Zr_2O^+ and UO^+ count-rate is also similar between the two minerals. We emphasize that uncertainty in the exact form of the relationship is subordinate to the high variation in U concentration observed in most samples and, because no suitably homogeneous standard mineral is available, no further attempt was made during this study to refine the relationship for baddeleyite. Absolute Th concentrations are determined from the U concentration and $^{232}Th/^{238}U$ ratio and are therefore subject to similar uncertainty.

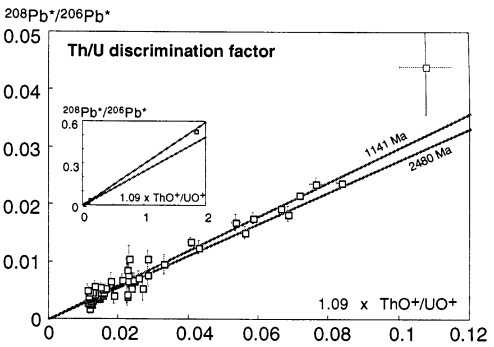


Fig. 6. Data used to calculate the $^{232}Th/^{238}U$ discrimination factor, *DF*. The diagram illustrates the agreement between $^{208}Pb^*/^{206}Pb^*$ versus $^{232}Th/^{238}U$ ($=1.09 \times ThO^+/UO^+$) and expected values (reference lines for three baddeleyite standards (details in Fig. 4) dated by isotope dilution methods). Measured $^{208}Pb/^{206}Pb$ ratios are corrected for common Pb using the 207-method.

Isotope ratios were determined by combining seven scans as a single analysis by using the linearity of count-rate for each mass with time as found for the great majority of analyses. Errors based on Poisson counting statistics constitute the minimum uncertainties in SHRIMP data. Scatter about each regression line in excess of counting statistics reflects target heterogeneity on the submicron scale and was incorporated by multiplying the uncertainty in the average count-rate of each mass by the square root of *F*, the ratio of the variance observed to that expected from ion counting statistics alone.

Within a single analytical session, relative $^{206}Pb^*/^{238}U$ ages of separate zircon analyses are assessed correctly using their individual errors as calculated theoretically, or as observed empirically within each analysis. When the mean $^{206}Pb^*/^{238}U$ age is reported for a group of analyses, we have added, in quadrature, the error (coefficient of variation) in the mean $^{206}Pb^*/^{238}U$ of the reference standard. Except where otherwise noted, all ages reported in this paper are quoted with 95% ($1\sigma \times$ Student's *t*) confidence limits.

5. Isotopic results

5.1. RL1 baddeleyite

During five separate analytical sessions (three using SHRIMP I and two using SHRIMP II), 137 analyses were obtained from 66 randomly oriented baddeleyite crystals. (Owing to the large size of this data set, it is not included here, but can be obtained on request from the first author.) U concentrations range from 50 to 950 ppm, averaging 250 ppm; Th content ranges from 1 to 84 ppm and averages 11 ppm; $^{232}Th/^{238}U$ ratios range from 0.01 to 0.17, with a mean of 0.04. The proportion of common ^{206}Pb to total ^{206}Pb is greater than 1% for only 7 of 137 analyses, so that in general, uncertainty due to common Pb correction is insignificant.

For Neoproterozoic and younger rocks, $^{207}Pb/^{206}Pb$ ages by both thermal ionization mass spectrometry and ion microprobe are determined with lower precision than those based on $^{206}Pb/^{238}U$ ratios. Errors on ages calculated from individual analyses reported here average ± 50 Ma (1σ); the low precision is a consequence of the relative insensitivity of the $^{207}Pb^*/^{206}Pb^*$ ratio as a measure of age in young samples, as well as the low abundance of radiogenic ^{207}Pb in young baddeleyite with average U concentration. However, the large number of analyses conducted allows a mean of high precision to be determined. Mean $^{207}Pb^*/^{206}Pb^*$ ratios for each of five sessions agree to within error and 137 analyses combined form a single, approximately Gaussian population (Fig. 7), with 94% of results falling within two standard deviations of the mean value of 0.06664 ± 0.00010, equivalent to 826.5 ± 3.1 Ma (1σ). The close agreement of all the baddeleyite $^{207}Pb^*/^{206}Pb^*$ ratios implies that Pb loss from the grains was either completely absent, or occurred recently if present. It is very unlikely that baddeleyite in mafic intrusions is of xenocrystic origin (Krogh et al., 1984; Heaman and LeCheminant, 1993), hence we regard 827 ± 6 Ma (2σ) to be the best estimate of the time of crystallization of the RL1 dyke.

5.2. RL1 zircon

Two analytical sessions on SHRIMP II were devoted to analysis of RL1 zircons. It was anticipated that the zircon results would complement those from RL1 baddeleyite, for which a $^{206}Pb^*/^{238}U$ age was not determined. Prior to session 2, the mount was repolished slightly and recoated with gold to ensure optimum conductivity. About two-thirds of the crystals measured during session 1 were re-analysed during the second session.

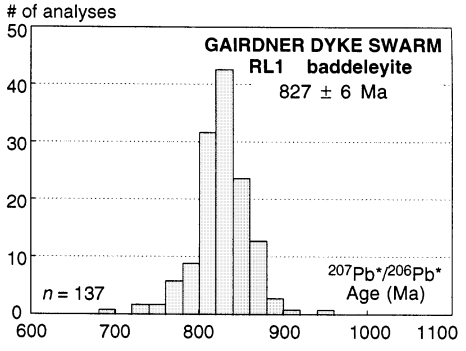


Fig. 7. Histogram of 137 $^{207}Pb^*/^{206}Pb^*$ ages for RL1 baddeleyite.

U–Th–Pb ratios and absolute abundances were determined relative to the SL13 standard zircon, analyses of which were interspersed with those of unknown grains. Although isotope-dilution methods indicate consistently an average $^{206}Pb^*/^{238}U$ age of 572 Ma for SL13 (Claoue-Long et al., 1995), extensive high spatial-resolution analysis using the SHRIMP ion microprobe confirms the presence of two age components. Using a maximum-likelihood mixture-modelling approach (Sambridge and Compston, 1994), an SL13 data-set of 235 analyses can be modelled well as two overlapping Gaussian distributions having mean ages of 578 Ma (64%) and 561 Ma, as determined by comparison with other well characterized zircons (W. Compston, pers. comm., 1996). These two components can be recognized in results from most analytical sessions and the factor necessary to equate the older measured component with 578 Ma can be used to adjust the $^{206}Pb^*/^{238}U$ ratios of unknown zircons. For RL1 zircons, this results in slight increases in $^{206}Pb^*/^{238}U$ ratios of 0.79% for the first session and 0.26% for the second, relative to those calculated by assuming a session mean of 572 Ma for SL13.

Measured U concentrations (Table 1) for normally-luminescent zircon in both analytical sessions vary from 170 to 1270 ppm, averaging ~ 600 ppm; Th ranges from 160 to 3000 ppm, with a mean of ~ 1250 ppm. Two analyses of non-luminescent zircon (10.1 and 14.1) yield significantly higher concentrations: ~ 3800 ppm U and ~ 9000 ppm Th. $^{232}Th/^{238}U$ ratios range from 0.58 to 6.8, with a mean of 2.1. The large range in these parameters is typical of igneous zircons from dolerite intrusions (e.g. Lanyon et al., 1993). Higher concentration values relative to coexisting baddeleyite are also typical (e.g. Krogh et al., 1984; Heaman et al., 1992; Romer et al., 1995) and suggest that baddeleyite crystallized before zircon, the latter forming in

Table 1
Ion microprobe U–Th–Pb analytical data for zircons from GDS dolerite sample RL1

Grain Area	CL	U (ppm)	Th (ppm)	Th/U (ppm)	f_{206} (%)	$^{207}Pb^*/^{206}Pb^*$ ($\pm\sigma$)	$^{206}Pb^*/^{238}U$ ($\pm\sigma$)	Apparent age (Ma $\pm 1\sigma$)
Session 1								
4.1	L	270	303	1.12	0.199	0.06564 ± 0.00093	0.1352 ± 0.0014	817.4 ± 8.1
2.1	L(+N?)	1008	2630	2.61	1.534	0.06479 ± 0.00059	0.1240 ± 0.0007	753.5 ± 4.1
*6.1	L	372	250	0.67	0.703	0.06259 ± 0.00095	0.1346 ± 0.0020	813.9 ± 11.4
7.1	L	998	1350	1.35	0.407	0.06599 ± 0.00038	0.1323 ± 0.0012	800.8 ± 7.0
8.1	L	919	893	0.97	0.110	0.06635 ± 0.00145	0.1283 ± 0.0037	777.9 ± 21.0
9.1	L	446	2029	4.55	0.464	0.06558 + 0.00059	0.1310 ± 0.0012	793.6 ± 6.9
*10.1	N	3948	8575	2.17	5.247	0.05976 ± 0.00072	0.1181 ± 0.0034	719.7 ± 19.8
11.1	L	1017	1608	1.58	0.046	0.06622 ± 0.00031	0.1331 ± 0.0008	805.2 ± 4.8
12.1	L	356	743	2.09	0.425	0.06499 ± 0.00097	0.1280 ± 0.0012	776.6 ± 7.1
13.1	L	171	162	0.95	1.020	0.08211 ± 0.02275	0.1304 ± 0.0038	790.2 ± 21.5
14.1	N	3785	9690	2.56	7.155	0.05782 ± 0.00098	0.1114 ± 0.0093	680.7 ± 54.0
15.1	L	671	996	1.48	0.302	0.06509 ± 0.00049	0.1351 ± 0.0008	817.1 ± 4.4
*16.1	L	415	291	0.70	6.527	0.01990 ± 0.00339	0.1558 ± 0.0053	986.3 ± 30.8
17.1	L	360	487	1.35	0.129	0.06543 ± 0.00075	0.1365 ± 0.0011	825.0 ± 6.4
18.1	L	1092	1430	1.31	0.214	0.06596 ± 0.00056	0.1218 ± 0.0011	740.9 ± 6.3
19.1	L	1298	2410	1.86	3.335	0.06516 ± 0.00085	0.1268 ± 0.0020	769.4 ± 11.2
*20.1	L	547	1393	2.55	0.420	0.06290 ± 0.00146	0.1364 ± 0.0011	824.4 ± 6.4
*21.1	L	334	2209	6.61	1.336	0.06322 ± 0.00102	0.1341 ± 0.0014	811.0 ± 7.9
22.1	L	261	271	1.04	0.040	0.18433 ± 0.00072	0.4954 ± 0.0062	2594 ± 27
*23.1	L	284	1204	4.25	1.326	0.06224 ± 0.00124	0.1365 ± 0.0011	825.0 ± 6.3
24.1	L	611	1219	1.99	0.514	0.06622 ± 0.00071	0.1308 ± 0.0009	792.3 ± 5.1
25.1	L	401	729	1.82	1.506	0.06610 ± 0.00108	0.1302 ± 0.0012	789.0 ± 6.7
26.1	L	869	3036	3.49	0.055	0.06623 ± 0.00036	0.1322 ± 0.0008	800.7 ± 4.3
27.1	L	633	1972	3.12	0.476	0.06532 ± 0.00074	0.1332 ± 0.0010	806.3 ± 5.6
22.2	L	786	15	0.02	0.195	0.15188 ± 0.00190	0.3903 ± 0.0047	2124 ± 22
Session 2								
19.2	L+N	1627	3164	1.95	4.363	0.06274 ± 0.00083	0.1249 ± 0.0029	758.9 ± 16.8
6.2	L	319	401	1.26	0.513	0.06332 ± 0.00099	0.1277 ± 0.0010	774.7 ± 5.7
4.2	L	291	334	1.15	0.059	0.06686 ± 0.00068	0.1375 ± 0.0016	830.4 ± 9.0
2.2	L+N	1294	3764	2.91	2.459	0.06401 ± 0.00064	0.1325 ± 0.0017	802.3 ± 9.8
11.2	L	996	1510	1.52	0.083	0.06569 ± 0.00033	0.1285 ± 0.0007	779.5 ± 3.8
26.2	L+N	931	3558	3.82	0.251	0.06597 ± 0.00034	0.1337 ± 0.0012	809.1 ± 6.6
17.2	L	385	531	1.38	0.010	0.06739 ± 0.00055	0.1356 ± 0.0013	819.4 ± 7.2
18.2	L	1189	1724	1.45	0.111	0.06607 ± 0.00027	0.1320 ± 0.0009	799.5 ± 5.0
31.1	L+N	1749	1012	0.58	1.107	0.06508 ± 0.00076	0.1439 ± 0.0083	866.8 ± 47.0
32.1	L	779	1679	2.16	2.203	0.06017 ± 0.00189	0.1486 ± 0.0150	893.1 ± 84.4
12.2	L(+N?)	725	1193	1.65	5.239	0.05194 ± 0.00316	0.1404 ± 0.0061	846.6 ± 34.3
*25.2	L	422	655	1.55	0.229	0.06621 ± 0.00059	0.1302 ± 0.0008	789.0 ± 4.6
20.2	L	851	2185	2.57	0.276	0.06380 ± 0.00101	0.1365 ± 0.0044	824.6 ± 25.2
21.2	L	337	2297	6.83	0.903	0.06394 ± 0.00135	0.1280 ± 0.0011	776.2 ± 6.4
*33.1	L+N	770	1642	2.13	1.179	0.06380 ± 0.00211	0.1573 ± 0.0040	941.5 ± 22.5
*34.1	L	586	1439	2.46	0.639	0.06437 ± 0.00063	0.1519 ± 0.0030	911.6 ± 16.7
*35.1	L	218	169	0.77	1.519	0.05647 ± 0.00561	0.1726 ± 0.0069	1026 ± 38
36.1	L	852	2462	2.89	2.441	0.06352 ± 0.00126	0.1360 ± 0.0021	822.1 ± 12.2
24.2	L+N	457	748	1.64	0.549	0.06507 ± 0.00076	0.1330 ± 0.0010	805.0 ± 5.4
37.1	L	232	208	0.90	0.293	0.06612 ± 0.00106	0.1424 ± 0.0014	858.2 ± 8.0
*38.1	L	461	327	0.71	1.918	0.06034 ± 0.00291	0.1540 ± 0.0057	923.1 ± 32.1
39.1	L	215	227	1.06	0.879	0.06350 ± 0.00170	0.1328 ± 0.0013	803.7 ± 7.3
7.2	L	1074	1640	1.53	0.125	0.06651 ± 0.00035	0.1362 ± 0.0016	823.3 ± 9.4
8.2	L	710	730	1.03	0.182	0.06595 ± 0.00069	0.1292 ± 0.0047	783.3 ± 26.7
*15.2	L	727	934	1.28	0.435	0.06419 ± 0.00067	0.1443 ± 0.0018	869.0 ± 10.0
23.2	L	344	1613	4.69	1.786	0.06518 ± 0.00130	0.1364 ± 0.0015	824.2 ± 8.4

magma more evolved and hence more enriched in silica and incompatible elements.

Common Pb content, estimated using non-radiogenic ^{204}Pb, is highly variable. Values for f_{206} (the proportion of common to total ^{206}Pb) range from much less than 1 to over 7% (Table 1). In Fig. 8, both uncorrected and 204-corrected compositions are plotted, so that the extent of correction using ^{204}Pb can be visualized. The highest proportions of common ^{206}Pb, 5.2 and 7.2%, were measured in non-luminescent zircon (analyses 10.1 and 14.1). High ^{204}Pb was also evident where the probe intersected cracks in the zircon. Several analyses in each session plot below Concordia, suggesting that ^{204}Pb has been overestimated and that Pb compositions for these analyses have been over-corrected. This interpretation is supported by mass scans that reveal a large, complex REE peak adjacent to the low-mass side of the ^{204}Pb peak, suggesting that for some analyses, a high-mass tail from high concentrations of REE caused a small isobaric interference under ^{204}Pb. We have used the 207-method (Compston et al., 1992, 1995; Roberts et al., 1995) to correct only six analyses in each session that are significantly reversely discordant when corrected using ^{204}Pb (details are given in Table 1).

In each session, there is a main group of analyses containing individual ^{206}Pb*/^{238}U ages ranging from about 830 to less than 700 Ma (Fig. 8); three analyses in session 1 and nine analyses in session 2, yield ^{206}Pb*/^{238}U ages ranging from about 860 to more than 1000 Ma. The observed distribution of apparent ages, after appropriate correction is made for common Pb, is illustrated in Fig. 9. Mixture-modelling reveals a series of age components for each session that correspond closely to the peaks observed in the probability distributions. Excluding, for the moment, results greater than 840 Ma, the oldest age components observed are 820.2 ± 4.5 Ma for session 1 and 822.7 ± 5.6 Ma for session 2 (both $\pm 1\sigma$ and including the error-component from age-calibration with respect to SL13). These components are within error of each other and within error of the mean age of RL1 baddeleyite at 826.5 ± 3.1 Ma (1σ), interpreted above as the age of crystallization of the RLI dyke. We therefore interpret the main group of analyses for each session as samples of a single population of magmatic zircons, in which the oldest age components have experienced little or no loss of radiogenic Pb.

This interpretation, however, is complicated by a few ^{206}Pb*/^{238}U ages (three in session 1 and 9 in session 2) that are reversely-discordant and range up to 1026 Ma. Based on the similarity of morphology, U and Th concentrations and alteration features with the rest of the sample, the apparently older zircons are unlikely to be xenocrysts. Cathodoluminescence images (Fig. 3) suggest that RL1 zircons were subjected to a hydrothermal or metamorphic fluid (possibly a late-stage, highly-fractionated magmatic liquid, e.g. Silver and Deutsch, 1963) that caused substantial and heterogeneous enrichment in U, Th, Pb and REE. Anomalously old and reversely-discordant ages could therefore reflect the presence of unsupported radiogenic Pb caused by internal isotopic redistribution, either by reaction with metasomatic fluids or via α-recoil, or both (Williams et al., 1984; Mattinson et al., 1996). Furthermore, corrosive fluids could selectively dissolve submicron-scale high-U metamict zones following internal redistribution, leaving a zircon enriched in low-U zones containing excess radiogenic Pb (Mattinson et al., 1996). It has been argued also that reversely-discordant ages for high-U zircons might result from radiation-induced microstruc-

Analyses are listed in the order in which they were conducted. CL refers to the appearance of the analysed area of each zircon using cathodoluminescence imaging: L, normally-luminescent; N, non-luminescent. All isotopic ratios and ages include correction for common Pb using ^{204}Pb except for ^{206}Pb*/^{238}U ratios and apparent ages for six analyses in each session (marked with *) that were corrected using the 207-method. f_{206} is the proportion of common ^{206}Pb to total measured ^{206}Pb, corrected using ^{204}Pb. Pb* denotes radiogenic Pb. The apparent age is calculated from ^{206}Pb*/^{238}U; decay constants used are those recommended by Steiger and Jager (1977). Two analyses (22.1, 22.2) in session 1 yield Archean ages, indicating that this zircon is either a xenocryst or a contaminant introduced during laboratory processing. ANU sample number 97-477.

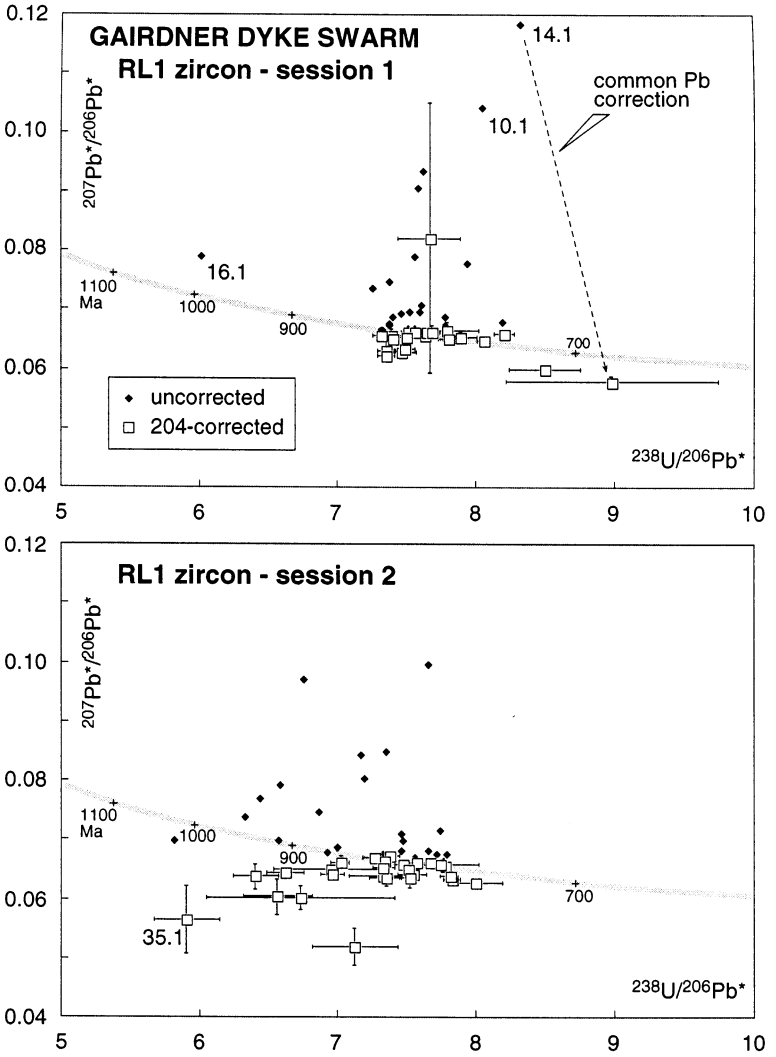


Fig. 8. U–Pb evolution diagrams illustrating SHRIMP analytical results for RL1 zircons. Error bars (1σ) are shown for 204-corrected data only. The dashed arrow in the top figure shows the trajectory of common Pb correction, along a mixing line between initial and radiogenic Pb. Analyses referred to in the text are labelled. The corrected composition for analysis 16.1 plots below the diagram.

tural changes that could enhance the sputtering of $^{206}Pb^{+}$ relative to $^{238}U^{+}$ during ion microprobe analysis (Black et al., 1991; McLaren et al., 1994). Further investigation of these interpretative options is beyond the scope of the present paper.

Regardless of the cause of the older apparent ages, we interpret the main group of analyses in each session to represent a single igneous population in which the zircons have lost variable amounts of radiogenic Pb. The oldest concordant age components in each session agree to within error with the baddeleyite $^{207}Pb^{*}/^{206}Pb^{*}$ age and thus strengthen further our interpretation of the baddeleyite age of 827 ± 6 Ma as an unambiguous age of emplacement for the RL1 dyke.

5.3. Little Broken Hill gabbro

Zircons and composite baddeleyite–zircon grains were analysed in sets of six scans during

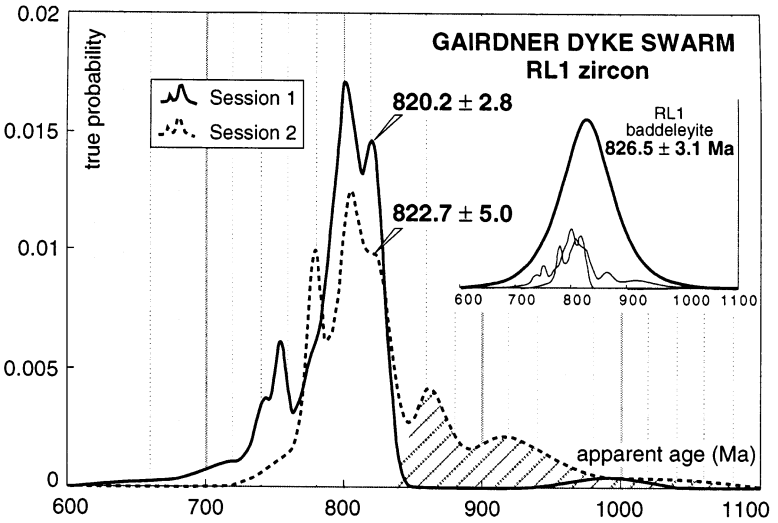


Fig. 9. Probability density plot illustrating the observed distribution of $^{206}Pb^{*}/^{238}U$ ages obtained for RL1 zircon. Excluding older analyses discussed in the text (indicated by diagonal pattern), the oldest age components for zircon (session 1: 820.2 ± 4.5 Ma; session 822.7 ± 5.6 Ma; both ±1σ, including SL13 calibration error) are in close agreement with the single-component peak for RL1 baddeleyite at 826.5 ± 3.1 Ma (1σ). Inset shows the (unstandardized) probability density curve for baddeleyite in relation to those for zircon.

one session on SHRIMP II. Isotopic ratios and U and Th concentrations for zircon were determined by comparison with the QGNG zircon standard, mounted with and analysed alternately with unknown zircons throughout the analytical session. QGNG zircons, from a quartz gabbronorite of the Lincoln Complex, South Australia (Mortimer et al., 1988), have been shown by high-precision isotope dilution analysis to have crystallized at 1850 Ma and to have remained largely undisturbed since that time (G.E. Mortimer, pers. comm., 1996). Mixture-modelling of $^{206}Pb^{*}/^{238}U$ ages from 17 different QGNG zircons obtained in this session found only a single age component, which was equated to 1850 Ma.

Isotopic ratios and concentrations for four baddeleyite analyses were also calculated relative to the QGNG standard (Fig. 10), using the power-law calibration exponent of 2.12 appropriate for baddeleyite, rather than the value of 2.00 used for zircon (Claoue-Long et al., 1995). Although the lower UO^{+}/U^{+} ratios for baddeleyite relative to the zircon standard reduce confidence in the calibration procedure and despite not taking into account any bias due to crystal orientation, the baddeleyite results can reasonably be considered

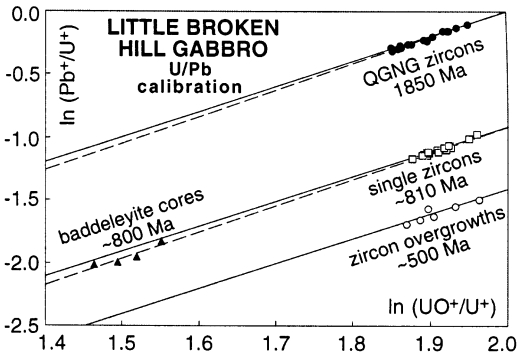


Fig. 10. Logarithmic calibration plot showing measured ion ratios for zircon and baddeleyite from the Little Broken Hill gabbro. Zircons are calibrated using a power law exponent of 2.00 (solid lines with slope = 2.00) as described by Claoue-Long et al. (1995); four baddeleyite analyses are calibrated against the QGNG zircon standard using an exponent of 2.12 (dashed lines).

as semi-quantitative estimates and are sufficient to support our arguments as laid out below.

Concentrations of U for single zircons range from 400 to 1100 ppm, with a mean of ~700 ppm, Th ranges from 300 to 2900 ppm, averaging ~1200 ppm and $^{232}Th/^{238}U$ ratios vary between 0.63 and 2.9, with a mean of 1.8. As noted earlier

for RL1 zircons, these values and their ranges, are typical of primary zircons from mafic intrusions. Zircon overgrowths on baddeleyite have lower values for all these parameters: U and Th average ~400 and 43 ppm, respectively; the low average $^{238}Th/^{238}U$ ratio of ~0.12 probably reflects growth of polycrystalline zircon by reaction with low $^{238}Th/^{238}U$ baddeleyite. Baddeleyite cores have an apparent U content between 100 and 750 ppm and $^{238}Th/^{238}U$ ratios between 0.02 and 0.07.

Common Pb was estimated using ^{204}Pb for analyses of single zircons; in all cases, f_{206} is much less than 1% (Table 2). Polycrystalline zircon overgrowths (with one exception) and baddeleyite cores are very low in $^{238}Th/^{238}U$ and were corrected for common Pb using the 208-method described above for RL1 baddeleyite. One overgrowth (C1.1) has higher $^{238}Th/^{238}U$ (0.54) and was corrected using ^{204}Pb. Values of f_{206} for baddeleyite cores range between 1.6 and 4.8% and for zircon overgrowths between 0.4 and 2.0%.

The $^{207}Pb^*/^{206}Pb^*$ ratios measured in 18 zircons agree within error and mixture-modelling finds only a single age component at 826.5 ± 4.5 Ma (1σ). The $^{206}Pb^*/^{238}U$ ratios (Fig. 11) are dispersed beyond analytical precision and yield a mean age of ~810 Ma, slightly discordant with respect to their mean $^{207}Pb^*/^{206}Pb^*$ age. Using mixture-modelling, the $^{206}Pb^*/^{238}U$ data can be resolved into an older population at 814.6 ± 2.9 Ma (67%) and a younger component with a mean of 801.9 ± 3.4 Ma (both $\pm 1\sigma$ and including error-component from QGNG). The older component, however, is still significantly younger than the mean $^{207}Pb^*/^{206}Pb^*$ age. We regard this difference to result from loss of radiogenic Pb and therefore consider the mean $^{207}Pb^*/^{206}Pb^*$ age of 826.5 ± 9.4 Ma to be the more reliable estimate of the age of these zircons.

Apparent $^{206}Pb^*/^{238}U$ ages for four baddeleyite cores are concordant within large errors and range from 785 to 826 Ma, in agreement with the single zircon results. Five of six zircon overgrowths analysed yield a concordant age of 499.6 ± 7.4 Ma. The excluded analysis (C3.1) yields an apparent $^{206}Pb^*/^{238}U$ age of 532 Ma, which suggests that the material analysed included a small proportion of primary baddeleyite in addition to zircon.

The agreement between baddeleyite and single zircon results is strong evidence that the two minerals were comagmatic. On this basis, we interpret the single zircons to be a sample from a magmatic population and regard the $^{207}Pb^*/^{206}Pb^*$ age of 827 ± 9 Ma to be the best estimate of the age of crystallization of the gabbro. Breakdown of baddeleyite to produce polycrystalline overgrowths of zircon occurred at 500 ± 7 Ma.

6. Discussion

6.1. Age of initial rifting in south-central Australia

The $^{207}Pb^*/^{206}Pb^*$ age of 827 ± 6 Ma inferred for emplacement of the RL1 dyke is indistinguishable from a zircon $^{207}Pb^*/^{206}Pb^*$ age of 824 ± 4 Ma for a dolerite dyke in the western Musgrave Inlier (Glikson et al., 1996), thus confirming the correlations between dykes in the two areas made previously on the basis of less precise Sm–Nd isochron ages and petrologic, stratigraphic and geochemical data (Goode, 1970; Crawford and Hilyard, 1990; Zhao and McCulloch, 1993; Zhao et al., 1994). These ages are also identical to the $^{207}Pb^*/^{206}Pb^*$ age of 827 ± 9 Ma obtained from zircon of the Little Broken Hill gabbro and support previous tentative correlations between the GDS and mafic intrusions in the Willyama Inier (Flint and Parker, 1993). Also within error of 827 Ma is a U–Pb upper-intercept age of 783 ± 42 Ma for conventionally-analysed zircon from the Boucaut Volcanics (Fanning, 1989 in Drexel et al., 1993) as well as an Rb–Sr isochron estimate of 830 ± 50 Ma for the Wooltana Volcanics (Compston et al., 1966).

Collectively, these results demonstrate the contemporaneity of most of the intrusive and volcanic rocks and establish firmly the areal extent of the Willouran Basic Province as defined by Crawford and Hilyard (1990). Because of the apparently higher propensity of baddeleyite, relative to zircon, to retain radiogenic Pb (Krogh et al., 1984; Heaman and LeCheminant, 1993) and the large number of analyses conducted (137) of RL1 baddeleyite, we consider the baddeleyite age of 827 ± 6 Ma to be the most accurate estimate of the

Table 2
Ion microprobe U–Th–Pb analytical data for zircon and baddeleyite from Little Broken Hill gabbro

Grain	U (ppm)	Th (ppm)	Th/U	f_{206} (%)	$^{207}Pb^*/^{206}Pb^*$ ($\pm\sigma$)	$^{206}Pb^*/^{238}U$ ($\pm\sigma$)	Apparent age (Ma $\pm 1\sigma$)
Single zircons							
1.1	804	1596	2.0	0.080	0.06657 ± 0.00047	0.1340 ± 0.0007	810.5 ± 4.2
2.1	803	1268	1.6	0.061	0.06654 ± 0.00050	0.1354 ± 0.0009	818.8 ± 4.9
3.1	724	1707	2.4	0.073	0.06695 ± 0.00068	0.1314 ± 0.0009	795.6 ± 4.9
4.1	772	1525	2.0	0.101	0.06576 ± 0.00048	0.1348 ± 0.0014	815.0 ± 8.0
5.1	1116	1819	1.6	0.316	0.06694 ± 0.00051	0.1370 ± 0.0013	827.6 ± 7.3
6.1	474	859	1.8	0.240	0.06576 ± 0.00097	0.1343 ± 0.0014	812.1 ± 8.2
7.1	586	818	1.4	0.060	0.06784 ± 0.00071	0.1320 ± 0.0014	799.2 + 7.8
8.1	673	1931	2.9	0.165	0.06693 ± 0.00059	0.1357 ± 0.0008	820.4 ± 4.7
9.1	472	707	1.5	0.140	0.06564 ± 0.00071	0.1347 ± 0.0013	814.7 ± 7.4
10.1	412	731	1.8	0.190	0.06689 + 0.00109	0.1335 ± 0.0015	807.8 + 8.8
11.1	799	1043	1.3	0.028	0.06742 ± 0.00046	0.1318 ± 0.0008	797.8 + 4.4
12.1	570	1250	2.2	0.138	0.06702 ± 0.00073	0.1324 ± 0.0009	801.4 ± 4.8
13.1	1115	2887	2.6	0.127	0.06611 ± 0.00045	0.1334 ± 0.0007	807.2 ± 4.0
14.1	547	660	1.2	0.163	0.06668 ± 0.00072	0.1344 ± 0.0013	813.0 ± 7.3
15.1	616	752	1.2	0.069	0.06651 ± 0.00067	0.1332 ± 0.0010	805.9 ± 5.5
16.1	607	1162	1.9	0.136	0.06672 ± 0.00060	0.1353 ± 0.0014	818.0 ± 7.7
17.1	545	1035	1.9	0.733	0.06604 ± 0.00145	0.1340 ± 0.0013	810.5 ± 7.2
18.1	492	309	0.63	0.280	0.06675 ± 0.00076	0.1352 ± 0.0011	817.4 ± 6.5
Baddeleyite cores							
C6.1	280	8	0.029	0.032	0.06336 ± 0.00402	0.1358 ± 0.0023	820.9 ± 12.9
C3.2	109	2	0.017	0.048	0.07568 ± 0.00565	0.1366 ± 0.0044	825.6 ± 24.9
C1.2	749	53	0.070	0.016	0.06737 ± 0.00153	0.1304 ± 0.0018	790.3 ± 10.2
C6.2	447	19	0.043	0.019	0.06626 ± 0.00238	0.1296 ± 0.0021	785.5 ± 11.8
Zircon overgrowths on baddeleyite							
C2.1	462	33	0.072	0.362	0.05734 ± 0.00100	0.0818 ± 0.0008	506.6 ± 4.9
C3.1	395	53	0.136	1.370	0.06119 ± 0.00132	0.0861 ± 0.0010	532.2 ± 5.7
C4.1	890	42	0.047	1.981	0.05781 ± 0.00124	0.0802 ± 0.0007	497.5 ± 3.9
C5.1	271	41	0.150	1.671	0.05822 ± 0.00190	0.0800 ± 0.0013	496.4 ± 7.7
C7.1	239	44	0.184	1.907	0.05756 ± 0.00145	0.0796 ± 0.0013	493.4 ± 7.7
*C1.1	337	183	0.542	0.941	0.05892 ± 0.00240	0.0807 ± 0.0012	500.2 ± 7.3

All data for single zircons are corrected for common Pb using ^{204}Pb; data for baddeleyite and zircon overgrowths are corrected for common Pb using the 208-method except analysis *C1.1, for which ^{204}Pb was used. Australian Geological Survey Organisation sample number 96 180 040. Refer to Table 1 for additional explanatory notes.

age of the mafic magmatic event and a minimum estimate for the base of the Adelaidean System.

The RL1 age is significantly older than the 802 ± 10 Ma age determined for the Rook Tuff (Fanning et al., 1986), a dacitic volcaniclastic unit that overlies the Wooltana Volcanics (Fig. 2). The felsic composition and the time difference of 25 Ma suggest that the Rook Tuff may be a product of crustal melting related to conductive heating above the Willouran plume (Campbell and Hill, 1988; Hill, 1991).

We interpret the breakdown of baddeleyite to form zircon in the Little Broken Hill gabbro to have occurred during metamorphism associated with the Delamerian Orogeny; uralitization of primary pyroxene at this time would have released silica necessary for the reaction to occur. In the Broken Hill region, late metamorphism was limited mainly to older shear zones reactivated during Delamerian deformation (Gibson et al., 1996, 1997). Rb–Sr and K–Ar ages of numerous biotite samples indicate that the Broken Hill region expe-

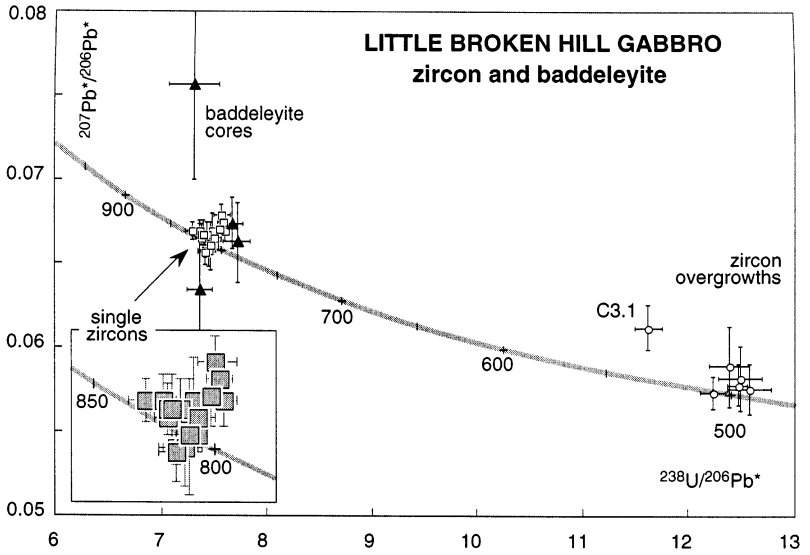


Fig. 11. U–Pb evolution diagram illustrating SHRIMP analytical results for zircon and composite baddeleyite–zircon crystals from the Little Broken Hill gabbro. Error bars are 1σ.

rienced a thermal pulse at ~520 Ma (Harrison and McDougall, 1981 and references therein). The $^{206}Pb^*/^{238}U$ age of 500 ± 7 Ma, inferred for formation of zircon overgrowths, is the most precise and, we suggest, the most reliable estimate of the age of this event in the Broken Hill region.

6.2. *Implications for the breakup of rodinia*

Park et al. (1995) proposed that three coeval mafic intrusive suites in western North America (inset, Fig. 1) represent sectors of a giant radiating dyke swarm fragmented by the separation of Australia and Laurentia and that the GDS is a missing sector of this swarm. Precise ages of the North America intrusions range from ~790 to ~760 Ma and are based on isotope-dilution U–Pb dating of baddeleyite and zircon and $^{40}Ar-^{39}Ar$ step-heating analysis of hornblende (references in Park et al., 1995). These intrusions are at least 40 Ma younger than the GDS and the Little Broken Hill gabbro. The time scale for formation of continental flood basalts and giant radiating dyke swarms that are related to mantle plumes is short, probably about 1 Ma and certainly less than 15 Ma (White and McKenzie, 1989; Hill et al., 1992); neither, hence the GDS nor the older mafic intrusive suite in the Willyama Inlier can be part of the same giant radiating dyke swarm as the ~780 Ma intrusions of western North America.

Based on paleomagnetic data and tectonostratigraphic analysis, Powell et al. (1993, 1994) argued that separation between Australia and Laurentia occurred at ~700 Ma. If this is correct, then neither the 827 Ma Willouran event nor the 780 Ma Laurentian event are likely to be related directly to continental breakup. Data from several younger CFB provinces suggest a link between continental breakup and the arrival of a starting plume; where initiation of a new spreading ridge follows flood volcanism, which marks the arrival of a new plume, the time interval is usually between 1 and 5 Ma, but can be as long as 40 Ma (Hill et al., 1992). Alternatively, the GDS and related rocks may represent an episode of entirely intracontinental extension (e.g. Siberian Traps, Columbia River events) and may be unrelated to breakup. In either case, intrusion of mafic magma at 827 Ma (or 780 Ma in North America) is likely to have induced thermal weakening of the lithosphere (Hill, 1991) and thus could have facilitated later continental separation driven by plate-scale, rather than plume-related, processes.

A link was suggested by Gibson et al. (1996,

1997) between continental breakup and the NW-trending dykes associated with the 827 Ma Little Broken Hill gabbro in the Willyama leer (Fig. 12). During continental breakup, coast-parallel dykes are emplaced perpendicular to the direction of extension, into thinned continental crust within 50 or 100 km of the margin, towards which they tend to increase in abundance (Nielsen, 1978; Hooper, 1990; Peate et al., 1990). Gibson et al. (1996, 1997) argued that the dykes in the Willyama Inier are situated close to the inferred Neoproterozoic margin of Australia, are associated with NW-trending Neoproterozoic normal faults (subsequently reactivated and overprinted extensively during Delamerian compression) and that both dykes and faults are parallel to the continental margin and perpendicular to the direction of breakup. If their hypothesis is correct, then breakup commenced by 827 Ma.

The orientation and distribution of the main part of the GDS (excluding dykes in the Willyama Inlier), however, are more consistent with emplacement above a mantle plume than with intrusion during extension associated with continental breakup. Dykes associated with a mantle plume form typically in a radial or fan pattern that can extend 2000 km or more from the plume centre (May, 1971; Fahrig, 1987; Ernst et al., 1993, 1995) and their orientation need bear no relation to a future continental margin or to the direction of breakup. The GDS was emplaced, in part, into relatively thick crust of the Gawler Craton, extends for more than 1000 km from the proposed margin into the centre of the continent and diverges slightly towards the north-west (Fig. 12). Dykes appear to be less numerous toward the continental margin (although within the Adelaide Geosyncline there are disrupted dolerites that could be GDS equivalents). Trace element ratios, Nd isotopic signatures and other geochemical characteristics are also consistent with a plume origin for the GDS (Crawford and Hilyard, 1990; Zhao et al., 1994).

It is possible that the GDS, together with the

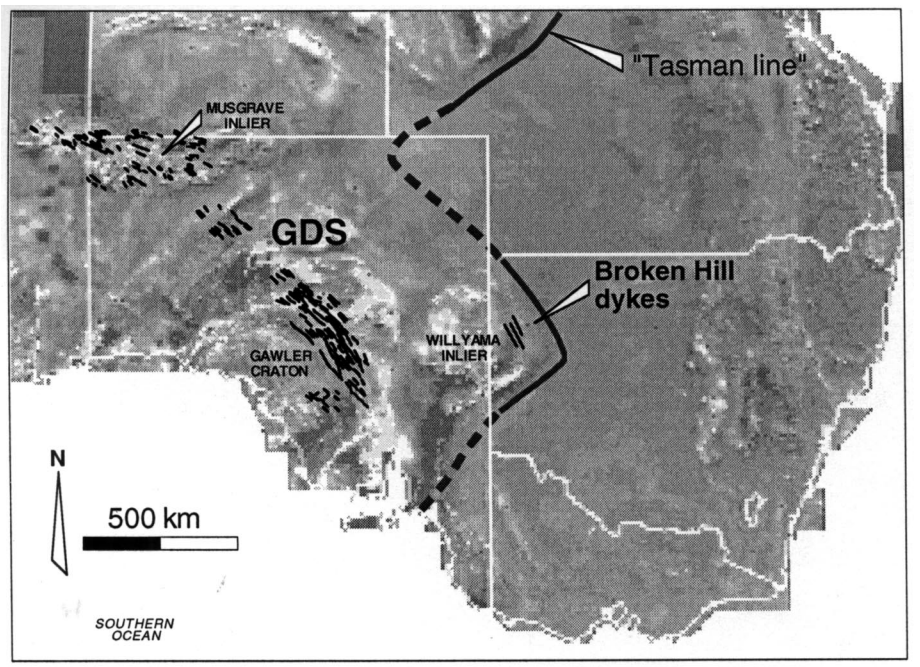


Fig. 12. Shaded-relief total magnetic intensity image of south-east Australia (courtesy of Australian Geological Survey Organization) showing the locations of the GDS and mafic dykes in the Willyama Inlier with respect to the inferred Neoproterozoic continental margin ('Tasman line' of Hill, 1951; Myers et al., 1996).

Adelaide Geosyncline, could represent a failed rift arm or aulacogen (Burke and Dewey, 1973; Rutland, 1973; Preiss, 1987). In this case, however, the rift geometry should conform roughly to that of a triple junction and breakup would have occurred and coast-parallel dykes would have been been emplaced along directions that need not be parallel to the GDS. No dykes of this age in suitable orientations have been identified, although if they were situated close to the continental margin, they may not have survived Delamerian tectonism. As noted above, the characteristics of the GDS more closely reflect emplacement above a mantle plume and therefore imply that the GDS was emplaced prior to breakup.

If rift-related extension in the Willyama Hier was underway by 827 Ma, then the identical age obtained for magmatism associated with the Willouran plume indicates that the two events were essentially synchronous (within the precision of the ages) at 827 Ma and hence it is likely that breakup was triggered by arrival of the Willouran plume.

Powell et al. (1993) argued, however, that apparent polar wander (APW) paths for Australia and Laurentia are coincident until ~725 Ma. Although paleomagnetic and baddeleyite U–Pb age data for the 723 Ma Franklin diabases and associated Natkusiak Basalts of northern Canada are precise and reliable (Fahrig et al., 1971; Heaman et al., 1992), the paleopole for East Gondwana at ~720 Ma employed by Powell et al. (1993) is based on less dependable data from mafic dykes of the south-western Yilgam Craton (YB dykes of Giddings, 1976) and the Malani Rhyolite of India (Klootwijk, 1975). The age of ~750 to ~700 Ma assigned originally to the YB dykes by Giddings (1976), was based on preliminary, unpublished Rb–Sr data of the late A.R. Crawford and should not be considered reliable (J.W. Giddings, W. Compston, pers. comm., 1996). In addition, the YB paleopole is of very low precision ($A_{95}=28.1^{\circ}$, Giddings, 1976). The Malani Rhyolite yielded a paleopole of higher precision (*dp* and *dm* = 8.0 and 11.5°, Klootwijk, 1975) and an Rb–Sr whole-rock isochron age of 729 ± 10 Ma (Crawford and Compston, 1970). The paleomagnetic results are apparently reliable, although the age should be treated with some caution given the difficulty, in general, of gauging the accuracy of Rb–Sr determinations.

In addition, the Tapley Hill Formation (Fig. 2), inferred by Powell et al. (1994) to represent the first major Neoproterozoic transgression of the Australian continent and to mark the inception of ocean-ridge spreading, was dated by Rb–Sr at 750 ± 53 Ma (Webb and Coats, 1980 in Preiss, 1987), using whole-rock shale samples now known to be unreliable for the purpose. Nevertheless, the precision of the age would allow deposition of the Tapley Hill Formation to be as early as 800 Ma. Finally, it should be noted that if initial continental separation was NE–SW-directed (present coordinates), as suggested by Scheibner (1973), Preiss (1987), Gibson et al. (1996, 1997) and Powell (1997), then a large component of motion was in a longitudinal sense with respect to early Neoproterozoic Australian and Laurentian paleopoles (Powell et al., 1993; Park et al., 1995) and would not be detected paleomagnetically. In view of these uncertainties and the arguments presented above, the possibility that breakup was significantly older than 725 Ma, perhaps older than 800 Ma, cannot be ruled out.

If the hypothesis that Australia was joined to Laurentia is correct, we would expect to see a complementary suite of 827 Ma dykes on the margins of Laurentia and/or 780 Ma dykes in Australia. If complementary dyke swarms existed, no evidence of them has been found, although both continental margins have been modified extensively by processes of continental breakup and terrane accretion. Although our results do not preclude correlation between Australia and Laurentia, they may lend support to the alternative hypothesis, suggested by Li et al. (1995, 1996) that the Yangtze and Cathaysia Cratons of South China were situated between Australia and Laurentia prior to the breakup of Rodinia. Rift-related volcanics dominate the Lower Sinian System along the northern and western margins of the Yangtze Craton and yield K–Ar ages between ~835 and 860 Ma (Liu, 1991; Li et al., 1995, 1996; Gao et al., 1996). Mafic intrusive rocks to the southeast, along an inferred Grenville-age suture with the Cathaysia Craton, yield ^{40}Ar–^{39}Ar and U–Pb

ages of 832±7 and 837 Ma, respectively (Liu, 1991; Zhou and Zhu, 1993; Li et al., 1995, 1996). Taking their ages at face value, these rocks appear to be roughly coeval with those of the 827 Ma Willouran event in Australia, although integrated geological, isotopic, paleomagnetic and geochemical studies of mafic intrusions in South China are required to test the correlation with confidence.

6.3. Future tests of the Rodinia hypotheses

Comparison of precisely synchronous paleomagnetic poles from different continents is the most reliable way to test (i.e. show consistency with or disprove conclusively) continental reconstruction hypotheses. In this case, the substantial difference in age between the GDS and the North American intrusions precludes this type of direct comparison. An alternative is to compare segments of APW paths that have been calibrated accurately by high-precision geochronology. However, such data for the Neoproterozoic of Australia and South China currently are insufficient to test the Rodinia hypotheses adequately (Idnurm and Giddings, 1988, 1995; Van der Voo and Meert, 1991; Li et al., 1996).

No paleomagnetic data for the GDS are yet available because most of the dyke swarm is not exposed at the surface. The WV1 paleopole for the Wooltana Volcanics (McWilliams and McElhinny, 1980), interpreted as primary by the original authors, is now thought to be an overprint acquired during Paleozoic hydrothermal activity (Idnurm and Heinrich, 1993; Powell et al., 1993; M. Idnurm, pers. comm., 1996). The new data presented in this paper indicate that dykes exposed on surface in the Musgrave and Willyama Inliers are similar in age to the GDS, hence paleomagnetic data for Australia at 827 Ma could be obtained from these rocks.

7. Conclusions

Our new data, in conjunction with previous and current isotopic, geochemical and stratigraphic information (Goode, 1970; Crawford and Hilyard, 1990; Zhao and McCulloch, 1993; Zhao et al., 1994; Glikson et al., 1996), confirm the synchronous nature of Willouran magmatism over most of south-central Australia. The best estimate of the age of this event and of initial rifting in the Adelaide Geosyncline and adjacent platforms, is 827±6 Ma, based on the mean $^{207}Pb^*/^{206}Pb^*$ age of 137 analyses of baddeleyite from dolerite of the GDS intersected in drill hole RL1. Partially metamict, high-uranium zircons from the same sample exhibit variable degrees of Pb loss, alteration and possibly inheritance. However, in each of two analytical sessions, the oldest concordant group of $^{206}Pb^*/^{238}U$ ages for zircon corresponds to the mean $^{207}Pb^*/^{206}Pb^*$ age for baddeleyite.

Analyses of zircon and baddeleyite indicate that the Little Broken Hill gabbro, emplaced during extension in the eastern Willyama Inlier, crystallized at 827±9 Ma, identical to the age inferred for emplacement of the GDS. In the same sample, breakdown of primary baddeleyite to form overgrowths of polycrystalline zircon occurred at 500±7 Ma, providing an improved estimate for the age, in this region, of the thermal event associated with the Delamerian Orogeny.

The GDS was emplaced at least 40 Ma before the ~780 Ma mafic intrusions in western North America studied by Park et al. (1995), hence the two cannot be part of the same giant radiating dyke swarm. If separation between Australia and Laurentia occurred at ~700 Ma, as proposed by Powell et al. (1994), then neither of these magmatic episodes is likely to be related directly to breakup, in which case breakup is likely to have been driven by plate scale, rather than plume-related, processes. If the Little Broken Hill gabbro and associated dykes in the Willyama Inlier are breakup-related, as argued by Gibson et al. (1996, 1997), then breakup commenced by 827 Ma and the Willouran mantle plume is likely to have played an active role in triggering the process. Our results show the GDS and related rocks in Australia to be roughly coeval with mafic igneous rocks in similar stratigraphic position in the Lower Sinian System of the Yangtze Craton and may support the hypothesis of Li et al. (1995, 1996) that parts of South China were situated between Australia and Laurentia prior to the breakup of Rodinia.

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References

Bell, R.T., Jefferson, C.W., 1987. An hypothesis for an Australian–Canadian connection in the Late Proterozoic and the birth of the Pacific Ocean. In: Proceedings of the Pacific Rim Congress '87, Parkville, Australia, Australian Institute of Mining and Metallurgy, Brisbane pp. 39–50.

Black, L.P., Kinny, P.D., Sheraton, J.W., 1991. The difficulties of dating mafic dykes: an Antarctic example. Contrib. Mineral. Petrol. 109, 183–194.

Borg, S.C., DePaolo, D.J., 1994. Laurentia, Australia and Antarctica as a Late Proterozoic supercontinent: constraints from isotopic mapping. Geology 22, 307–310.

Boyd, D.M., Tucker, D.H., 1990. Australian magnetic dykes. In: Parker, A.J., Rickwood, P.C., Tucker, D.H. (Eds.), Mafic Dykes and Emplacement Mechanisms. Balkema, Rotterdam, pp. 391–399.

Brookfield, M.E., 1993. Neoproterozoic Laurentia–Australia fit. Geology 21, 683–686.

Buchan, K.L., Halls, H., 1990. Paleomagnetism of Proterozoic mafic dyke swarms of the Canadian Shield. In: Parker, A.J., Rickwood, P.C., Tucker, D.H. (Eds.), Mafic Dykes and Emplacement Mechanisms. Balkema, Rotterdam, pp. 209–230.

Burke, K., Dewey, J.F., 1973. Plume-generated triple junctions: key indicators in applying plate tectonics to old rocks. J. Geol. 81, 406–433.

Campbell, I.H., Hill, R.I., 1988. A two-stage model for the formation of the granite–greenstone terrains of the Kalgoorlie–Norseman area Western Australia. Earth Planet. Sci. Lett. 90, 11–25.

Claoue-Long, J.C., Compston, W., Roberts, J., Fanning, C.M., 1995. Two Carboniferous ages: a comparison of SHRIMP zircon ages with conventional zircon ages and $^{40}Ar/^{39}Ar$ analysis. In: Geochronology, Time Scales, and Global Stratigraphic Correlation, SEPM Special Publication, vol. 54, pp. 3–21.

Compston, W., Crawford, A.C., Bofinger, V.M., 1966. A radiometric estimate of the duration of sedimentation in the Adelaide Geosyncline, South Australia. Geol. Soc. Aust. J. 13, 229–276.

Compston, W., Kinny, P.D., Williams, I.S., Foster, J.J., 1986. The age and Pb 1055 behaviour of zircons from the Isua supracrustal belt as determined by ion microprobe. Earth Plan. Sci. Lett. 80, 71–81.

Compston, W., Williams, I.S., Meyer, C., 1984. U–Pb geochronology of zircons from lunar breccia 73217 using a sensitive high mass-resolution ion microprobe. J. Geophys. Res. Suppl. 89, B525–B534.

Compston, W., Williams, I.S., Kirschvink, J.L., Zhang, Z., Ma, G., 1992. Zircon U–Pb ages for the Early Cambrian time scale. J. Geol. Soc. Lond. 149, 171–184.

Compston, W., Sambridge, M.S., Reinfrank, R.F., Moczydlowska, M., Vidal, G., Claesson, S., 1995. Numerical ages of volcanic rocks and the earliest faunal zone within the Late Precambrian of east Poland. J. Geol. Soc. Lond. 152, 599–611.

Cowley, W.M., Flint, R.B., 1993. Epicratonic igneous rocks and sediments. In: Drexel, J.F., Preiss, W.V., Parker, A.J. (Eds.), The Geology of South Australia, vol. 1. The Precambrian, S. Aust. Geol. Surv. Bull. 54 (1993) 142–147.

Crawford, A.J., Hilyard, D., 1990. Geochemistry of Late Proterozoic flood basalts, Adelaide Geosyncline, South Australia. Geol. Soc. Aust. Spec. Publ. 16, 49–67.

Crawford, A.R., Compston, W., 1970. The age of the Vindhyan System of Peninsular India. Q. J. Geol. Soc. Lond. 125, 351–371.

Dalziel, I.W.D., 1991. Pacific margins of Laurentia and East Antarctica–Australia as a conjugate rift pair: evidence and implications for an Eocambrian supercontinent. Geology 19, 598–601.

Dalziel, I.W.D., 1992. Antarctica: A tale of two supercontinents? Ann. Rev. Earth Planet. Sci. 20, 501–526.

Dalziel, I.W.D., Lawver, L.A. 1995. Plumes and the dynamics of supercontinental fragmentation. 1995 Spring Meeting, Abstract Volume, supplement to EOS, Transactions of the American Geophysical Union, p. F172.

Daziel, I.W.D., 1997. Where stands Rodinia and Panotia? In: Bradshaw, J.D. and Weaver, S.D. (Eds.), Terrane Dynamics 97, Abstr. Int. Conf. Terrane Geology, Christchurch, NZ, pp. 52–53.

Drexel, J.F., Preiss, W.V., Parker, A.J. (Eds), The Geology of South Australia, vol. 1, Precam. S. Aust. Geol. Surv. Bull. 54 (1993) 242.

Eisbacher, G.H., 1985. Late Proterozoic rifting, glacial sedimentation, and sedimentary cycles in the light of Windermere deposition, western Canada. Palaeogeog. Palaeoclim. Palaeoecol. 51, 231–254.

Ernst, R.E., Head, J.W., Parfitt, E., Wilson, L., Grosfills, E., 1993. Giant radiating dyke swarms on Earth and Venus. Lunar Planet. Sci. Conf. XXIV, 447–448.

Ernst, R.E., Buchan, K.L., Palmer, H.C., 1995. Earth's largest dyke swarms: their characteristics, distribution and importance in solving geological problems. In: Baer, G., Heimann, E. (Eds.), Physics and Chemistry of Dykes. Balkema, Rotterdam, p. 321.

Fahrig, W.F., Irving, E., Jackson, G.D., 1971. Paleomagnetism of the Franklin diabases. Can. J. Earth Sci. 8, 455–467.

Fahrig, W.F., 1987. The tectonic settings of continental mafic dyke swarms: Failed arm and early passive margin. In: Halls, H.C., Fahrig W.F. (Eds.), Mafic Dyke Swarms. Geological Association of Canada, Special Paper no. 34. pp. 331–348.

Fanning, C.M., Flint, R.B., Preiss, W.V., 1983. Geochronology of the Pandurra Formation. Geol. Surv. S. Aust. Q. Geol. Notes 88, 11–16.

Fanning, C.M., Ludwig, K.R., Forbes, B.G., Preiss, W.V., 1986. Single and multiple grain U–Pb zircon analyses for the early Adelaidean Rook Tuff, Willouran Ranges, South Australia. Abstr. Geol. Soc. Aust. 15, 71–72.

Flint, R.B., Daly, S.J., 1993. Coompana Block. In: Drexel, J.F., Preiss, W.V., Parker, A.J. (Eds.), The Geology of South Australia, vol. 1, The Precambrian. S. Aust. Geol. Surv. Bull. 54, 168–169.

Flint, R.B., Parker, A.J., 1993. Willyama Inliers. In: Drexel, J.F., Preiss, W.V., Parker, A.J. (Eds.), The Geology of South Australia, vol. 1, The Precambrian. S. Aust. Geol. Surv. Bull. 54, 82–93.

Gao, S., Zhang, B.-R., Wang, D.-P., Ouyang, J.-P., Xie, Q.-L., 1996. Geochemical evidence for the Proterozoic tectonic evolution of the Qinling Orogenic belt and its adjacent margins of the North China and Yangtze Cratons. Precambrian Res. 80, 23–48.

Gibson, G.M., Maidment, D.W., Haren, R., 1996. Re-evaluating the structure of Broken Hill. Aust. Geol. Surv. Org. Res. Newslett. 25, 1–3.

Gibson, G.M., Maidment, D.W., Haren, R., 1997. Willyama Supergroup, Broken Hill, Australia: a 1600 Ma granulite terrane situated along the Neoproterozoic margin of Gondwana following continental rifting and the breakup of rodinia. In: Bradshaw, J.D., Weaver, S.D. (Eds.), Terrane Dynamics 97, Abstracts of the International Conference on Terrane Geology, Christchurch, NZ, pp. 71–74.

Giddings, J.W., 1976. Precambrian paleomagnetism in Australia I: basic dykes and volcanics from the Yilgarn Block. Tectonophysics 30, 91–108.

Giles, C.W., Teale, G.S., 1979. A comparison of the geochemistry of the Roopena Volcanics and the Beda Volcanics. Geol. Surv. S. Aust. Q. Geol. Notes 71, 7–13.

Glikson, A.Y., Stewart, A.J., Ballhaus, C.G., Clarke, G.L., Feeken, E.H.J., Leven, J.H., Sheraton, J.W., Sun, S.-S., 1996. Geology of the western Musgrave Block, central Australia, with particular reference to the mafic-ultramafic Giles Complex. Aust. Geol. Surv. Org. Bull. 239, 206

Goode, A.D.T., 1970. The petrology and structure of the Kalka and Ewarara layered basic intrusions, Giles Complex, central Australia. Ph.D thesis, University of Adelaide.

Harrison, T.M., McDougall, I., 1981. Excess ^{40}Ar in metamorphic rocks from Broken Hill, New South Wales: implications for $^{40}Ar/^{39}Ar$ age spectra and the thermal history of the region. Earth Planet. Sci. Lett. 55, 123–149.

Heaman, L.M., LeCheminant, A.N., Rainbird, R.H., 1992. Nature and timing of Franklin igneous events, Canada: implications for a Late Proterozoic mantle plume and the breakup of Laurentia. Earth Planet. Sci. Lett. 109, 117–131.

Heaman, L.M., LeCheminant, A.N., 1993. Paragenesis and U–Pb systematics of baddeleyite (ZrO_2). Chem. Geol. 110, 95–126.

Hill, D., 1951. Geology. In: Handbook of Queensland. Australian Association for the Advancement of Science, Brisbane, pp. 13–24.

Hill, R.I., 1991. Starting plumes and continental breakup. Earth Planet. Sci. Lett. 104, 398–416.

Hill, R.I., Campbell, I.H., Davies, G.F., Griffiths, R.W., 1992. Mantle plumes and continental tectonics. Science 256, 186–193.

Hilyard, D., 1990. Willouran Basic Province: Stratigraphy of Late Proterozoic flood basalts, Adelaide Geosyncline, South Australia. Geol. Soc. Aust. Spec. Publ. 16, 34–48.

Hinthome, J.R., Andersen, C.A., Conrad, R.L., Lovering, J.F., 1979. Single-grain, $^{207}Pb^*/^{206}Pb^*$ and U/Pb age determinations with a 10 μm spatial resolution using the ion microprobe mass analyzer (IMMA). Chem. Geol. 25, 271–303.

Hoffman, P.F., 1991. Did the breakout of Laurentia turn Gondwanaland inside-out? Science 252, 1409–1412.

Hooper, P.R., 1990. The timing of crustal extension and the eruption of continental flood basalts. Nature 345, 246–249.

Idnurm, M., Giddings, J.W., 1988. Australian Precambrian polar wander: a review. Precambrian Res. 40, 61–88.

Idnurm, M., Giddings, J.W., 1995. Paleoproterozoic–Neoproterozoic North America Australia link: new evidence from paleomagnetism. Geology 23, 149–152.

Idnurm, M., Heinrich, C.A., 1993. A paleomagnetic study of hydrothermal activity and uranium mineralization at Mt Painter, South Australia. Aust. J. Earth Sci. 40, 87–101.

Jefferson, C.W., 1978. Correlation of Middle and Upper Proterozoic strata between north-western Canada and south and central Australia. Geol. Soc. Am. Programs Abstr. 7, 429

Klootwijk, C.T., 1975. A note on the paleomagnetism of the late Precambrian Malani Rhyolites near Jodhpur. India J. Geophys. 41, 189–200.

Krogh, T.E., Davis, D.W., Corfu, F., 1984. Precise U–Pb zircon and baddeleyite ages for the Sudbury area. In: Pye, E.G., Naldrett, A.J. and Giblin, P.E. (Eds.), The Geology and Ore Deposits of the Sudbury Structure, Chapter 20. Public Services Sector, Government of Ontario, Toronto, pp. 431–446.

Krogh, T.E., Corfu, F., Davis, D.W., Dunning, G.R., Heaman, L.M., Kamo, S.L., Machado, N., 1987. Precise U–Pb isotopic ages of diabase dykes and mafic to ultramafic rocks using trace amounts of baddeleyite and zircon. Geol. Assoc. Can. Spec. Pap. 34, 147–152.

Lanyon, R., Black, L.P., Seitz, H., 1993. U-Pb zircon dating of mafic dykes and its application to the Proterozoic geological history of the Vestfold Hills, East Antarctica. Contrib. Mineral. Petrol. 115, 184–203.

Li, Z.X., Zhang, L., Powell, C.McA., 1995. South China in Rodinia: part of the rrtissing link between Australia–East Antarctica and Laurentia? Geology 23, 407–410.

Li, Z.X., Zhang, L., Powell, C.McA., 1996. Positions of the East Asian cratons in the Neoproterozoic supercontinent Rodinia. Aust. J. Earth Sci. 43, 593–604.

Liu, H.Y., 1991. The Sinian System in China. Science Press, Beijing, p. 380 (in Chinese).

Lofgren, C., 1980. Experimental studies on the dynamic crystallization of silicate melts. In: R.B. Hargraves (Ed.), Physics of Magmatic Processes. Princeton University Press, USA, pp. 487–551.

Mattinson, J.M., Graubard, C.M., Parkinson, D.L., McClelland, W.C., 1996. U–Pb reverse discordance in zircons: the role of fine-scale oscillatory zoning and sub-micron transport of Pb. In: Basu, A., Hart, S. (Eds.), Earth processes: reading the isotopic code. American Geophysical Union, Geophysics Monograph 95, American Geophysical Union, Washington, D.C. pp. 355–370.

May, P.R., 1971. Pattern of Triassic–Jurassic diabase dykes around the North Atlantic in context of predrift position of the continents. Geol. Soc. Am. Bull. 82, 1285–1292.

McLaren, A.C., Fitz Gerald, J.D., Williams, I.S., 1994. The microstructure of zircon and its influence on the age determination from Pb/U isotopic ratios measured by ion microprobe. Geochim. Cosmochim. Acta 58, 993–1005.

McMenamin, M.A., Schulte-McMenamin, D.L., 1990. The Emergence of Animals: The Cambrian Breakthrough. Columbia University Press, New York, p. 217.

McWilliams, M.O., McElhinny, M.W., 1980. Late Precambrian paleomagnetism of Australia: the Adelaide Geosyncline. J. Geol. 88, 1–26.

Moores, E.M., 1991. Southwest US–East Antarctic (SWEAT) connection: a hypothesis. Geology 19, 425–428.

Morgan, W.J., 1981. Hotspot tracks and the opening of the Atlantic and Indian Oceans. In: C. Emiliani (Ed.), The Sea, vol. 7. Wiley Interscience, New York, pp. 443–487.

Mortimer, G.E., Cooper, J.A., Oliver, R.L., 1988. The geochemical evolution of Proterozoic granitoids near Port Lincoln in the Gawler Orogenic Domain of South Australia. Precambrian Res. 4041, 387–406.

Myers, J.S., Shaw, R.D., Tyler, I.M., 1996. Tectonic evolution of Proterozoic Australia. Tectonics 15, 1431–1446.

Nielsen, T.D.F., 1978. The Tertiary dyke swarm of the Kangerdlugssuaq area, East Greenland. An example of magmatic development during continental breakup. Contrib. Mineral. Petrol. 67, 63–78.

Park, J.K., Buchan, K.L., Harlan, S.S., 1995. A proposed giant radiating dyke swarm fragmented by the separation of Laurentia and Australia based on paleomagnetism of ca. 780 Ma mafic intrusions in western North America. Earth Planet. Sci. Lett. 132, 129–139.

Peate, D.W., Hawkesworth, C.J., Mantovani, M.S.M., Shukowsky, W., 1990. Mantle plumes and flood-basalt stratigraphy in the Parana, South America. Geology 18, 1223–1226.

Powell, C.McA., 1997. Assembly and breakup of Rodinia in Australia. European Union of Geosciences, EUG 9, Abstracts, European Union of Geosciences, Strasbourg 164.

Powell, C.McA., Li, Z.X., McElhinny, M.W., Meert, J.G., Park, J.K., 1993. Paleomagnetic constraints on timimg of the Neoproterozoic breakup of Rodinia and the Cambrian formation of Gondwana. Geology 21, 889–892.

Powell, C.McA., Preiss, W.V., Gatehouse, C.G., Krapez, B., Li, Z.X., 1994. South Australian record of a Rodinian epicontinental basin and its mid-Neoproterozoic breakup ($\sim$700 Ma) to form the Palaeo-Pacific Ocean. Tectonophysics 237, 113–140.

Preiss, W.V. (Compiler), 1983. Adelaide Geosyncline and Stuart Shelf: Precambrian and Paleozoic geology (with special reference to the Adelaidean). South Australian Department of Mines and Energy, 1:600 000 scale map.

Preiss, W.V., 1987. The Adelaide Geosyncline–Late Proterozoic stratigraphy, sedimentation, paleontology, and tectonics. Geol. Surv. S. Aust. Bull. 53, 438.

Preiss, W.V., 1990. A stratigraphic and tectonic overview of the Adelaide Geosyncline, South Australia. Geol. Soc. Aust. Spec. Publ. 16, 1–33.

Preiss, W.V., 1993. Neoproterozoic. In: Drexel, J.F., Preiss, W.V., Parker, A.J. (Eds.), The Geology of South Australia, vol. 1, The Precambrian. S. Aust. Geol. Surv. Bull. 54, 171–203.

Roberts, J., Claoue-Long, J.C., Jones, P.J., Foster, C.B., 1995. SHRIMP zircon age control of Gondwanan sequences in Late Carboniferous and Early Pern-dan Australia. In: Dunay, R.E., E.A. Hailwood (Eds.), Non-biostratigraphical Methods of Dating and Correlation. Geol. Soc. London, 89, 145–174.

Romer, R.L., Scharer, U., Wardle, R.J., Wilton, D.H.C., 1995. U–Pb age of the Seal lake Group, Labrador: relationship to Mesoproterozoic extension-related magmatism of Laurasia. Can. J. Earth Sci. 32, 1401–1410.

Ross, G.M., Parrish, R.R., Winston, D., 1992. Provenance and U–Pb geochronology of the Middle Proterozoic Belt Supergroup (north-western United States): implications for the age of deposition and pre-Panthalassa plate reconstructions. Earth Planet. Sci. Lett. 113, 57–76.

Rutland, R.W.R., 1973. Tectonic evolution of the continental crust of Australia. In: Tarling, D.H., Runcom, S.K. (Eds.), Implications of Continental Drift to the Earth Sciences, vol. 2, Academic Press, London, pp. 1011–1033.

Sambridge, M.S., Compston, W., 1994. Mixture modelling of multi-component data sets with application to ion-probe zircon ages. Earth Planet. Sci. Lett. 128, 373–390.

Scheibner, E., 1973. A plate tectonic model of the Paleozoic tectonic history of New South Wales. J. Geol. Soc. Aust. 20, 405–426.

Shimizu, N., Hart, S.R., 1982. Isotope fractionation in secondary ion mass spectrometry. J. Appl. Phys. 53, 1303–1311.

Silver, L.T., Deutsch, S., 1963. Uranium–lead isotopic variations in zircons: a case study. J. Geol. 71, 721–758.

Silverman, B.W., 1986. Density Estimation for Statistics and Data Analysis. Chapman and Hall, London, p. 175.

Steiger, R.H., Jager, E., 1977. Subcommission on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth Planet. Sci. Lett. 36, 359–362.

Stroud, W.J., Willis, I.L., Bradley, G.M., Brown, R.E., Stevens, B.P.J., Barnes, R.G., 1983. Rocks of the Broken Hill Group: their classification, nature, stratigraphic distribution, and origin. Geol. Surv. New S. Wales Rec. 21, 227–288.

Tucker, D.H., Boyd, D.M., 1987. Dykes of Australia detected by airborne magnetic surveys. In: Halls, H.C., Fahrig, W.F. (Eds.), Mafic Dyke Swarms. Geol. Assoc. Can. Spec. Pap. 34, 163–172.

Van der Voo, R., Meert, J.G., 1991. Late Proterozoic paleomagnetism and tectonic models: a critical appraisal. Precambrian Res. 53, 149–163.

Webb, A.W., Horr, G., 1978. The Rb–Sr age and petrology of a flow from the Beda Volcanics. Geol. Surv. S. Aust. Q. Geol. Notes 66, 10–13.

White, R., McKenzie, D., 1989. Magmatism at rift zones: the generation of volcanic continental margins and flood basalts. J. Geophys. Res. 94, 7685–7729.

Williams, I.S., Compston, W., Black, L.P., Ireland, T.R., Foster, J.J., 1984. Unsupported radiogenic Pb in zircon: a cause of anomalously high Pb–Pb, U–Pb, and Th–Pb ages. Contrib. Mineral. Petrol. 88, 322–327.

Wingate, M.T.D., 1997. Testing Precambrian Continental Reconstructions Using Ion Microprobe U–Pb Baddeleyite Geochronology and Paleomagnetism of Mafic Igneous Rocks. Ph.D thesis, Australian National University.

Wingate, M.T.D., Compston, W., Campbell, I.H., 1995. Testing the 'Vaalbara' hypothesis using combined ion microprobe U–Pb baddeleyite geochronology and paleomagnetism. 1995 GAC/MAC Annual Meeting, Program with Abstracts, p. A-111.

Young, G.M., 1992. Late Proterozoic stratigraphy and the Canada–Australia connection. Geology 20, 215–218.

Zhao, J.-X., McCulloch, M.T., 1993. Sm-Nd mineral isochron ages of Late Proterozoic dyke swarms in Australia: evidence for two distinct events of mafic magmatism and crustal extension. Chem. Geol. 109, 341–354.

Zhao, J.-X., McCulloch, M.T., Korsch, R.J., 1994. Characterization of a plume-related ~800 Ma magmatic event and its implications for basin formation in central-southern Australia. Earth Planet. Sci. Lett. 121, 349–367.

Zhou, X., Zhu, Y., 1993. Late Proterozoic collisional orogen and geosuture in south-eastern China: petrological evidence. Chin. J. Geochem. 12, 239–251.